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Total Least Squares Fitting of k -Spheres in n -D Euclidean Space Using an $(n + 2)$ -D Isometric Representation

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Keywords sphere fitting, hypersphere fitting, circle fitting, total least squares, hyperaccuracy, geometric algebra, conformal geometric algebra, Minkowski metric, horosphere

Abstract We fit k -spheres optimally to n -D point data, in a geometrically total least squares sense. A specific practical instance is the optimal fitting of 2D-circles to a 3D point set.

Among the optimal fitting methods for 2D-circles based on 2D (!) point data compared in [Al-Sharadqah & Chernov 2009], there is one with an algebraic form that permits its extension to optimally fitting k -spheres in n -D. We embed this ‘Pratt 2D circle fit’ into the framework of conformal geometric algebra (CGA), and doing so naturally enables the generalization. The procedure involves a representation of the points in n -D as vectors in an $(n + 2)$ -D space with attractive metric properties. The hypersphere fit then becomes an eigenproblem of a specific symmetric linear operator determined by the data. The eigenvectors of this operator form an orthonormal basis representing perpendicular hyperspheres. The intersection of these are the optimal k -spheres; in CGA the intersection is a straightforward outer product of vectors.

The resulting optimal fitting procedure can easily be implemented using a standard linear algebra package; we show this for the 3D case of fitting spheres, circles and point pairs. The fits are optimal (in the sense of achieving the KCR lower bound on the variance).

We use the framework to show how the hyperaccurate fit hyperspheres of [Al-Sharadqah & Chernov 2009] is a minor rescaling of the Pratt fit hypersphere.

1 Introduction

Fitting circles to planar point data has been studied extensively (see the references in [1]). The sum of least squares distances of N points at locations $\mathbf{p}_i$ to a circle with center $\mathbf{c}$ and radius ρ is

$$\sum_i \left(\sqrt{(\mathbf{p}_i - \mathbf{c}) \cdot (\mathbf{p}_i - \mathbf{c})} - \rho \right)^2. \quad (1)$$

This is algebraically much more awkward to optimize than a line fit based on the least square error of the same point data to a line with normal $\mathbf{n}$ and distance δ to the origin:

$$\sum_i \left(\mathbf{p}_i \cdot \mathbf{n} - \delta \right)^2. \quad (2)$$

Exact least squares optimization of eq.(1) leads to expensive iterative methods, whereas eq.(2) can be solved almost directly, as an eigenproblem of the direction of the vectors relative to the centroid of the data (which can be proved to lie on the optimal line).

Algebraic methods have been introduced to approximate the ‘exact’ solution of eq.(1) in a computationally acceptable non-iterative manner. To illustrate some of the variability of the results of such methods, Figure 1 shows a plane with 6 points (data from [12]), with a selection of ‘best fit’ circles determined by various methods from literature (we will briefly discuss their rationales in Section 4). Until very recently, no comparison had been made of the accuracy of the various approximations. Then in 2009, Al-Sharadqah & Chernov [1] showed some amazing results: when the

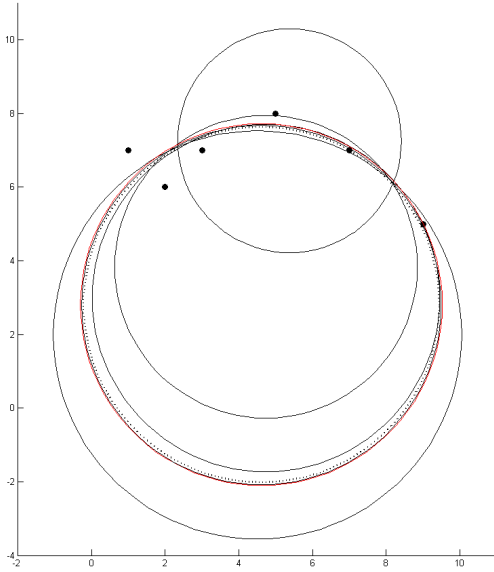


Fig. 1 Point data from [12] and the ‘best fit’ circles according to (bottom to top along the mid-line): Nievergelt [16]; Pratt (and this paper) [18] in red; Taubin [21] (virtually indistinguishable from Pratt at this resolution); the hyperaccurate fit from [1] (dashed); the geometric total least squares fit of eq.(1) (iteratively determined by [12]); Coope [4]; Gander’s algebraic fit [12]. Although no ground truth is known for this data, the hyperaccurate fit (dashed) is statistically best motivated. Note that the Pratt fit used in this paper is very close to it (a fact we will explain in Section 7), and that both are close to the geometric total least squares fit. More simulations may be found the Appendix.

accuracy is measured in terms of the mean square error (MSE) for data with a radial standard deviation of σ_{radial} from the circular contour, all ‘sensible’ unbiased circle fits are the same in their leading error term, which is of order $\sigma_{\text{radial}}^2/N$, and which moreover achieves the optimal Kanatani-Cramér-Rao lower bound for this problem. The subtle difference in MSE quality between the various fits that had been proposed is a term of order σ_{radial}^4 , which they call the *essential bias*, and which only manifests itself with enough data. Moreover, and most surprisingly, the seemingly desirable geometric total least squares estimator based on eq.(1) is actually asymptotically biased. They also used their analysis to design a new algebraic ‘hyperaccurate’ fit with zero essential bias (The ideas behind this estimator subsequently led to the ‘hyper-renormalization’ technique in [14], applicable in more general contexts than circle fitting). Since [1], we know that it is possible to have a non-iterative algebraic method that gives better circles than the iterative geometric total least squares optimization would yield.

Limiting ourselves to the estimators from [1], we used their Table 2 to establish the weighting factors of the two MSE terms of variance and essential bias. For uniformly distributed points along a unit circle with $\sigma_{\text{radial}}/\rho = 0.05$, the essential bias term is as large as the variance term for $N \approx 1000$ (for the ‘Pratt fit’, the others are of similar order of magnitude). The differences between the fits, and the hyperaccuracy of the new fit, therefore manifest themselves only when there is an excessive amount of point data. For a more reasonable amount, all ‘sensible’ estimators of the type studied in [1] are virtually identical in MSE quality, and all optimal in achieving the KCR lower bound. We show some simulations illustrating these points from [1] in Appendix A.

A circle is a 1-sphere; the point data resides in 2-D. The results from [1] generalize simply to the fitting of $(n-1)$ -spheres in n -D. But what if we want to find the best 2-D circles (1-spheres) to fit 3D spatial data, or, more generally, k -spheres in n -D?

In this paper we show that one of the circle fits studied in [1] has an algebraic form that permits this extension to k -spheres in a fairly straightforward manner, once an effectively chosen representational framework is employed. It is the fit originally introduced by Pratt in 1987 [18], which in [1] is given as the classical form of its cost function in terms of the coordinates (x_i, y_i) of $\mathbf{p}_i$ and a characterization of the circle by algebraic parameters (A, B, C, D) :

$$\sum_i \frac{(A(x_i + y_i)^2 + Bx_i + Cy_i + D)^2}{B^2 + C^2 - 4AD}. \quad (3)$$

We employ a fairly recent algebraic framework, *conformal geometric algebra* (CGA) [3, 15, 8], in which this cost function in n -D is formulated as

$$\sum_i \frac{(p_i \cdot x)^2}{x^2}, \quad (4)$$

with p_i a specific non-linear representation of the data point at location $\mathbf{p}_i$, and x a specific vector representation of the hypersphere to be found. We show how to solve this fit in closed form, retrieving the Pratt fit as the solution to a pure eigenproblem (rather than a constrained eigenproblem as in [1]). We view k -spheres as intersections of hyperspheres, and use the power of CGA to represent these as subspaces of its algebra. Since they are elements of computation (e.g., hypercircles are ‘factorizable bivectors’), we can formulate and solve the optimal k -sphere fit as an eigenproblem. We relate the solution to the full set of eigensolutions of the hypersphere fit problem already solved - geometrically, these represent a basis of orthogonal spheres. This provides a straightforward, practical formula to produce an

optimal k -sphere fit in n -D. We demonstrate how it can be implemented in a standard linear algebra package.

We then show that the framework thus constructed for the Pratt fit can be used, with one minor modification, to compute the hyperaccurate fit from [1]. In the process, we prove that the hyperaccurate fit has the same center as the Pratt fit (as may be seen in Figure 1).

This paper is structured in a tutorial fashion, as follows: Section 2 is a brief introduction into the CGA representation that enables our analysis, focused on its k -sphere representations. Section 3 derives the eigenproblem for the Pratt fit for hyperspheres, and shows how to implement it for spheres in 3D. We discuss the geometric nature of this algebraic fit when comparing it to some of the other methods in Section 4. Section 5 introduces the bivector representation of hypercircles (i.e. $(n-2)$ -spheres) in n -D, and solves their best fit (in the Pratt sense). We derive the relationship between hypersphere fitting and hypercircle fitting, and apply this to circles in 3D. Section 6 gives the general framework for k -sphere Pratt fitting in n -D as an extension of the observed hypercircle/hypersphere correspondence. Section 7 establishes the embedding of the hyperaccurate hyperspheres into CGA, and discusses its extension to k -spheres. Section 8 briefly indicates how the fitting of k -D flats can be performed by CGA. Section 9 draws some conclusions.

2 Representation of Rounds and Flats in Conformal Geometric Algebra

As we indicated, we make the fitting problems tractable by choosing a natural algebraic representation for the problem of hypersphere fitting. Geometrically, there are three types of elements that play a role in this problem: *points* (as the way the data is given), *hyperspheres* (as the desired regular solution) and *hyperplanes* (as possible degenerate solutions).

An ‘algebra of hyperspheres’ would be a natural representation, since points are simply hyperspheres of zero radius, and hyperplanes are spheres with an infinite radius. So we want a space in which n -D hyperspheres are represented as vectors; if we do so in a homogeneous manner (where multiples of vectors represent the same geometrical element), this space should have $(n+2)$ dimensions. We should design this space to have a sensible metric, so that its dot product is immediately meaningful to computing Euclidean distances in our original problem.

Conformal geometric algebra (CGA) fulfills these specifications. It uses an $(n+2)$ -dimensional metric

space to represent the (hyper)spheres, (hyper)planes and points. Its metric is the indefinite Minkowski metric; any orthonormal basis for it has one vector with a negative unit norm. This is unfamiliar, and perhaps unexpected in the clearly Euclidean problem of total least squares sphere fitting.¹ Like all geometric algebras (or Clifford algebras), CGA provides the capability of directly computing with its subspaces; since those subspaces represent intersections of spheres, the desired extension of the representation to hypercircles and hyperlines, or more general ‘rounds’ and ‘flats’, is automatically included.²

Because conformal geometric algebra is not very familiar, we give a brief overview of its construction, focusing on the properties relevant to our treatment of least squares k -sphere fitting. We therefore specify explicitly how it models Euclidean (rather than conformal) geometry.

– Setting up the representation space

Embed Euclidean n -space in an $n+2$ -dimensional space, with two extra dimensions. Introduce an orthonormal basis in this $(n+2)$ -D space, with n unit vectors $\mathbf{e}_i$ ($i = 1, \dots, n$) for the Euclidean part, and the vectors e_+ and e_- for the two extra dimensions. (Throughout, we will denote in **bold** the purely Euclidean elements, belonging to the embedded n -D Euclidean subspace.) Define a dot product according to Table 2(a): all basis vectors have zero dot product with basis vectors other than themselves; only $e_- \cdot e_- = -1$, all other vectors ‘square to +1 under the dot product’. Because of having this indefinite metric with $(n+1)$ basis vectors squaring to +1 and one vector squaring to -1 , this representational metric space is denoted as $\mathbb{R}^{n+1,1}$.

It is customary to introduce a different basis, no longer orthonormal: retain the Euclidean $\mathbf{e}_i$ but use $n_o = \frac{1}{2}(e_- + e_+)$ and $n_\infty = e_- - e_+$. It is easy to verify that the dot product table of this new basis is as Table 2(b). We use this table as the metric matrix $[M]$, for the *null basis* $\{\mathbf{e}_i, n_o, n_\infty\}$. The name ‘null

¹ Actually, the use of an $(n+2)$ -D space to represent n -D Euclidean space in a homogeneous, isometric manner is not a new trick. Wachter suggested it in 1816 in a letter to his teacher Gauss, and Lobachevsky called it a ‘horosphere’, but it has lain dormant. It was recently revived in the context of studying Euclidean geometry as a special case of conformal geometry by means of geometric algebra [3] [15].

² *Geometric algebra* is a framework to compute metrically with subspaces of a vector space, and with linear transformations on them. *Conformal geometric algebra* sets up an $(n+2)$ -D vector space that conveniently represents the conformal (i.e., angle preserving) transformations in n -D, by means of its geometric algebra; hence its name. Among those conformal transformations are the Euclidean similarities, which is why CGA is suitable for Euclidean problems of a metric nature.

$\cdot$	$\mathbf{e}_1$	$\mathbf{e}_2$	$\cdots$	$\mathbf{e}_n$	e_+	e_-
$\mathbf{e}_1$	1	0	$\cdots$	0	0	0
$\mathbf{e}_2$	0	1	$\cdots$	0	0	0
$\vdots$	$\vdots$	$\vdots$	<ddots< math=""></ddots<>	$\vdots$	0	0
$\mathbf{e}_n$	0	0	$\cdots$	1	0	0
e_+	0	0	$\cdots$	0	1	0
e_-	0	0	$\cdots$	0	0	-1

(a)

$\cdot$	$\mathbf{e}_1$	$\mathbf{e}_2$	$\cdots$	$\mathbf{e}_n$	n_o	n_∞
$\mathbf{e}_1$	1	0	$\cdots$	0	0	0
$\mathbf{e}_2$	0	1	$\cdots$	0	0	0
$\vdots$	$\vdots$	$\vdots$	<ddots< math=""></ddots<>	$\vdots$	0	0
$\mathbf{e}_n$	0	0	$\cdots$	1	0	0
n_o	0	0	$\cdots$	0	0	-1
n_∞	0	0	$\cdots$	0	-1	0

(b)

Fig. 2 Defining the metric of the $(n+2)$ -dimensional representation space through specifying the dot product of basis vectors, see text.

basis' reflects the fact that $n_o \cdot n_o = 0$ and $n_\infty \cdot n_\infty = 0$; note that $n_o \cdot n_\infty = -1$. The space is still called $\mathbb{R}^{n+1,1}$, since the choice of a different basis does not affect its metric structure.

– Points, distances

A simple way to characterize the essence of CGA is to display how a vector p of $\mathbb{R}^{n+1,1}$ represents a (weighted) point of the Euclidean space at a location $\mathbf{p}$:

$$p = \alpha (\mathbf{p} + n_o + \frac{1}{2} \|\mathbf{p}\|^2 n_\infty). \quad (5)$$

We will only need to consider unit-weight points in this paper, for which $\alpha = -n_\infty \cdot p = 1$. The element n_o is seen to represent the unit weight point at the origin (take $\mathbf{p} = \mathbf{0}$), the element n_∞ can be shown to represent the point at infinity. We will refer to such a vector p representing a point as a 'conformal point'.

With this point representation, one may compute the squared distance of two unit-weight points at finite Euclidean locations $\mathbf{p}$ and $\mathbf{q}$ from the corresponding conformal points p and q as their dot product:

$$\begin{aligned} p \cdot q &= (\mathbf{p} + n_o + \frac{1}{2} \|\mathbf{p}\|^2 n_\infty) \cdot (\mathbf{q} + n_o + \frac{1}{2} \|\mathbf{q}\|^2 n_\infty) \\ &= \mathbf{p} \cdot \mathbf{q} - \frac{1}{2} \|\mathbf{p}\|^2 - \frac{1}{2} \|\mathbf{q}\|^2 \\ &= -\frac{1}{2} \|\mathbf{p} - \mathbf{q}\|^2. \end{aligned} \quad (6)$$

Therefore the dot product in the space $\mathbb{R}^{n+1,1}$ can be interpreted as a squared Euclidean distance in the original Euclidean n -D target space. As a consequence, distance preserving transformations in Euclidean space can be represented as orthogonal transformations of the conformal space $\mathbb{R}^{n+1,1}$. (There is

some additional subtlety in the treatment of non-unit-weight points, leading to general conformal transformations, but we will not need those for this paper.) Note that this dot product is translation invariant, in contrast to the dot product in a classical homogeneous coordinate approach.

Another feature of eq.(6) is that the *squared* Euclidean distance appears as a (bi-)linear expression in the representative space. That makes squared distance considerations more tractable algebraically than distances *per se*, and we will use this feature in our hypersphere fitting.

Eq.(6) implies that $\|p\|^2 = p \cdot p = 0$ for a conformal vector p representing a Euclidean point. Such a p is called a *null vector*. It is the Minkowski metric of the representational space $\mathbb{R}^{n+1,1}$ that permits the existence of nonzero null vectors.

– Implementation

Bearing in mind our application, let us show that the point representation and the isometric dot product are easily implemented using a standard linear algebra package. On a basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n, n_o, n_\infty\}$, a point at Euclidean location $\mathbf{p}$ is represented by the unit-weight vector $[p]^T = [\mathbf{p}^T, 1, \frac{1}{2} \|\mathbf{p}\|^2]^T$, and the dot product $p \cdot q$ by employing the metric matrix $[M]$ taken from Figure 2 (with $[1]$ the $n \times n$ identity matrix):

$$\begin{aligned} p \cdot q &\leftrightarrow [p]^T [M] [q] \\ &= [\mathbf{p}^T \ 1 \ \frac{1}{2} \|\mathbf{p}\|^2] \begin{bmatrix} [1] & \mathbf{0} & \mathbf{0} \\ \mathbf{0}^T & 0 & -1 \\ \mathbf{0}^T & -1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ 1 \\ \frac{1}{2} \|\mathbf{q}\|^2 \end{bmatrix} \\ &= [\mathbf{p}^T \ 1 \ \frac{1}{2} \|\mathbf{p}\|^2] \begin{bmatrix} \mathbf{q} \\ -\frac{1}{2} \|\mathbf{q}\|^2 \\ -1 \end{bmatrix} \\ &= [\mathbf{p}^T \mathbf{q} - \frac{1}{2} \|\mathbf{q}\|^2 - \frac{1}{2} \|\mathbf{p}\|^2] \\ &= [-\frac{1}{2} \|\mathbf{p} - \mathbf{q}\|^2]. \end{aligned}$$

– Hyperspheres

A hypersphere s with as center a unit weight point c (at Euclidean location $\mathbf{c}$) and a radius of ρ is represented by the conformal space vector:

$$s = c - \frac{1}{2} \rho^2 n_\infty, \quad (7)$$

(or any multiple of this). Note that $-n_\infty \cdot s = -n_\infty \cdot c = 1$ (so this is a 'unit weight sphere' with n_o -coordinate equal to 1). Also, $\|s\|^2 = s \cdot s = \rho^2$, so s is not a null vector unless $\rho = 0$. A point can be viewed as a hypersphere of zero radius.

This representation of a hypersphere as a vector s should be interpreted as: a point p is on the hypersphere s if and only if $p \cdot s = 0$. (In this sense, s is a 'dual' representation of the hypersphere, the points

on it belong to its ‘inner product null space’ [17].) The proof is straightforward:

$$0 = p \cdot s = p \cdot c - \frac{1}{2} \rho^2 p \cdot n_\infty = -\frac{1}{2} (\|p - c\|^2 - \rho^2),$$

equivalent to the purely Euclidean equation of a hypersphere $\|p - c\|^2 = \rho^2$. This justifies the identification of s in eq.(7) as a (dual) sphere.

Two real Euclidean spheres intersect perpendicularly iff $\|c_1 - c_2\|^2 = \rho_1^2 + \rho_2^2$; this translates neatly into the algebraic orthogonality condition $s_1 \cdot s_2 = 0$ for their conformal representatives (and vice versa). A vector of the form $s_- = c + \frac{1}{2} \rho^2 n_\infty$ represents an ‘imaginary’ sphere with radius ρ and center c , since it is obtained from eq.(7) by replacing ρ^2 by $-\rho^2$. A real sphere s is algebraically orthogonal to an imaginary sphere s_- (in the sense that $s \cdot s_- = 0$) if it geometrically intersects s_- in a great circle.

– Hyperplanes

A hyperplane with unit normal vector $\mathbf{n}$ and passing through a point q is represented by the conformal vector:

$$\pi = \mathbf{n} + (q \cdot \mathbf{n}) n_\infty, \quad (8)$$

(or a multiple thereof). One easily verifies that $p \cdot \pi = 0$ for a conformal point p if and only if $(p - q) \cdot \mathbf{n} = 0$, which is the correct condition in purely Euclidean terms. Note that $n_\infty \cdot \pi = 0$, this is geometrically interpreted as: the point at infinity is on every hyperplane. Note also that $\|\pi\|^2 = \pi \cdot \pi = \mathbf{n} \cdot \mathbf{n} = \|\mathbf{n}\|^2 \neq 0$, so π is not a null vector.

– Subspaces as Blades; Outer Product, Scalar Product, Inner Product

More extended elements denoting higher dimensional algebraic subspaces are constructed using the ‘outer product’ of vectors (a product from Grassmann algebra also available in geometric algebra). On vectors, the outer product (denoted $\wedge$) is a bilinear anti-symmetric product and results in an element called a *2-blade*. Since $a \wedge b = -b \wedge a$ for all vectors a and b , it follows that $a \wedge a = 0$.

The outer product can be extended to other elements of the geometric algebra in a linear and associative manner, and then produces a *k-blade* when used on k vector factors (though the result may be zero). A scalar is a 0-blade, and $\alpha \wedge X = \alpha X$ for scalar α and any element X . A *k-blade* is the algebraic representation of a k -dimensional subspace of the vector space. Its dimensionality k is referred to as its *grade*.

Between elements of the same grade, a *scalar product* denoted $*$ is defined, producing a scalar as an extension of the metric dot product defined for vectors. The scalar product of two elements A and B is

symmetric: $A * B = B * A$. For elements of different grade, one can extend the dot product to an inner product, by the implicit definition

$$(A \wedge B) * C = A * (B \cdot C). \quad (9)$$

In practice, it is more convenient to define the inner product recursively. The inner product of a vector x and a k -blade is then an extension of the dot products of its vector factors:

$$x \cdot (a \wedge A) = (x \cdot a) \wedge A - a \wedge (x \cdot A). \quad (10)$$

(The outer product with a scalar coincides with scalar multiplication.) Similar extensions can be made to the inner product of k -blades and ℓ -blades, recursively using the defining identity

$$(B \wedge b) \cdot X = B \cdot (b \cdot X). \quad (11)$$

Together eq.(10) and eq.(11) ultimately express an inner product of blades in terms of the inner products of a vector factorization. Such a factorization always exists, and may be constructed by a Gram-Schmidt type algorithm.

These products can be extended in bilinear fashion to sums of blades, even of different grades (such elements are called *multivectors*).³

– Intersection, hyperlines, hypercircles

The intersection of two elements (dually) represented by conformal vectors is dually represented as their outer product.

As a consequence, the circle that is the intersection of a sphere s and a plane π is (dually) represented as the 2-blade $\kappa = s \wedge \pi$. This is easy to motivate, as follows. Rewriting the condition $p \cdot \kappa = 0$ for a point p to be on the circle κ , we find by eq.(10):

$$0 = p \cdot \kappa = p \cdot (s \wedge \pi) = (p \cdot s) \pi - (p \cdot \pi) s. \quad (12)$$

Since the vectors s and π are linearly independent, this is equivalent to ‘ $p \cdot s = 0$ and $p \cdot \pi = 0$ ’, so that to be on κ , a point should indeed be both on the sphere and on the plane. We can also characterize the circle as the product of two spheres $\kappa = s_1 \wedge s_2$ - though this representation is of course also not unique.

– Bivectors versus 2-Blades

The 2-blades formed as the outer product of vectors are of grade 2. Such elements of grade 2 can be considered as a vector space (denoted as $\bigwedge^2 \mathbb{R}^{n+1,1}$). It

³ For readers familiar with alternatives, in all formulas in this paper the notation ‘ $\cdot$ ’ signifies an inner product called the *left contraction* (see [8]) if used between multivectors with unequal grades; it coincides with the scalar product when used between equal grade multivectors.

is then natural to consider forming the sum of 2-blades. However, an element obtained in this manner is in general not a 2-blade, since it cannot necessarily be factored as the outer product of two vectors. It is called a *bivector*. The necessary and sufficient condition for a bivector X to be a 2-blade is

$$X \wedge X = 0 \quad (13)$$

(see [10]). Thus the 2-blades lie on a quadratic manifold within the linear space of bivectors. In 2D and 3D, all bivectors are 2-blades. Similarly, k -blades form a linear space in which one can construct k -vectors. More extended conditions than eq.(13) apply for k -vectors to be k -blades, see [10]. Only k -blades represent k -dimensional subspaces of the vector space of the algebra. In CGA, these subspaces can represent round or flat elements of Euclidean geometry.

– Geometric product

The main structural contribution of geometric algebra is the definition of a *geometric product* between vectors, which is bilinear, associative and invertible. The invertibility property allows compact notation and solution of equations involving geometrical elements, often in a coordinate-free manner.⁴

Although one can take the geometric product as basic and derive inner product and outer product from it, for this paper it is convenient to turn this around, and define the geometric product aB of a vector a and a multivector B as:

$$aB = a \cdot B + a \wedge B. \quad (14)$$

Note that $aa = a \cdot a$ is a scalar. The geometric product of a scalar α with a multivector B is just αB , and scalars commute: $\alpha B = B\alpha$. Geometrical relationships like proportionality (or ‘parallelness’) and perpendicularity of vectors become algebraic properties like commutativity and anti-commutativity of their geometric product.

The inverse of a non-null vector x is $x^{-1} = x/(x^2) = x/(x \cdot x)$. The inverse of a non-null 2-blade X is $X^{-1} = X/X^2 = X/(X * X)$.

– Multivector differentiation

As in classical vector calculus, differentiation with respect to a vector x is a vector operator (classically denoted by ∇_x , we use ∂_x following [13]). With the geometric product and the multivector structure of geometric algebra, this naturally extends to differentiation with respect to elements of the algebra [13, 7].

⁴ In Clifford algebra, this geometric product is called the Clifford product.

In this paper, we need only a few elementary results of multivector differentiation, combined with a product rule for the differentiation of products:

$$\partial_X (AB) = \dot{\partial}_X A B + \partial_X A \dot{B}, \quad (15)$$

where the overdot denotes what is being differentiated (the non-commutativity of the geometric product precludes moving the differentiation operator next to its argument).

$$\partial_X (X * A) = A \quad (16)$$

$$\partial_X (X * X)^{-1} = -2X(X * X)^{-2} \quad (17)$$

Here A is a multivector of the same grade as X . Such results may be found in [13].

3 Hypersphere Fitting

We are now ready to represent our hypersphere fitting problem in terms of the conformal geometric algebra of $\mathbb{R}^{n+1,1}$.

3.1 An Approximate Least Squares Criterion

The location vector $\mathbf{p}_i$ of a point p_i can be characterized relative to a hypersphere with center $\mathbf{c}$ and radius ρ as $\mathbf{p}_i = \mathbf{c} + (\rho + \delta_i) \mathbf{e}$, for an appropriately chosen unit vector $\mathbf{e}$. The signed distance δ_i between point and hypersphere is computed from this as

$$\delta_i = \sqrt{(\mathbf{p}_i - \mathbf{c}) \cdot (\mathbf{p}_i - \mathbf{c})} - \rho. \quad (18)$$

An optimal hypersphere to a set of points can then be defined as the one that minimizes the sum of squared differences $\sum_i \delta_i^2$. As we already mentioned in the introduction, the resulting cost criterion is an awkward expression in the hypersphere parameters, and a closed form solution has not been found; the typical solution is an iterative optimization using a gradient of the cost function.

Using the CGA representation, we can approximate the squared signed distance by an algebraically more convenient expression, involving the unit-weight conformal point $p_i = n_o + \mathbf{p}_i + \frac{1}{2}\mathbf{p}_i^2 n_\infty$ and the sphere $s = c - \frac{1}{2}\rho^2 n_\infty$, with c the unit-weight conformal point at location $\mathbf{c}$. For we have:

$$\begin{aligned} (p_i \cdot s)^2 / s^2 &= \frac{1}{4}(\rho^2 - \|\mathbf{p}_i - \mathbf{c}\|^2)^2 / \rho^2 \\ &= \frac{1}{4}(\rho^2 - 2\delta_i\rho - \delta_i^2 - \rho^2)^2 / \rho^2 \\ &= \frac{1}{4}(-\delta_i(2\rho + \delta_i))^2 / \rho^2 \\ &= \delta_i^2 \left(1 + \frac{\delta_i}{2\rho}\right)^2 \\ &= \delta_i^2 \left(1 + \frac{\delta_i}{\rho} + \left(\frac{\delta_i}{2\rho}\right)^2\right) \\ &\approx \delta_i^2, \end{aligned} \quad (19)$$

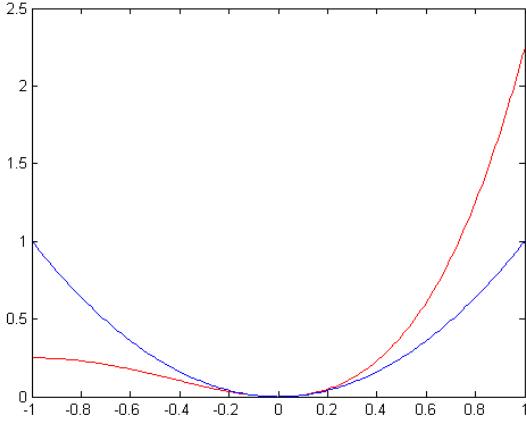


Fig. 3 Approximated squared distance (the red parabola) compared to the exact squared distance of points (the blue curve), as a function of radial distance along a line through the center, with 0 denoting the location of the spherical surface; the distance unit is rescaled by ρ , so that -1 denotes the center.

with a relative error of $(\delta_i/\rho) + \frac{1}{4}(\delta_i/\rho)^2$. The advantage of this approximation to the true squared distance is that its derivative is algebraically much more easily manipulable than the exact expression above – and that will lead to a closed form solution for the optimal hypersphere under the associated criterion. Note that the expression is invariant to a scalar weight of the vector x representing the hypersphere. Figure 3 shows a graph of this approximated square distance.

Even for points close to the hypersphere, Figure 3 shows that the approximate distance is asymmetric: it slightly overestimates the distance for points outside, and underestimates for inside, relative to the exact squared distance. It is therefore not a surprise that this measure might produce a slightly biased fit; the big surprise is that [1] has shown that minimizing the exact squared distance as in eq.(1) is a biased overestimate as well! The intuitive explanation is that points inside and outside the hypersphere do not pull the hypersphere in exactly the same amount, due to their slightly different density. The hyperfit of [1] corrects this, by adding a counteracting data-dependent asymmetry to the point contributions. We will explain this modification in Section 7; for now, let us work with the simpler measure $(p_i \cdot s)^2/s^2$ and solve its fits within CGA. Immediately afterward, we show in Section 4.1 that our distance measure is actually equivalent to an algebraic constraint on the sphere fit by Pratt [18]; but we need the CGA formulation as a springboard for the extension to k -sphere fits (which is the main contribution of this paper).

3.2 Optimal Hypersphere Fitting as an Eigenproblem

With the approximate squared distance criterion above, our estimation problem becomes:

Find x that minimizes: $\mathcal{L}(x) = \frac{1}{N} \sum_i (p_i \cdot x)^2/x^2$. (20)

For an optimum, we demand that the derivative $\partial_x \mathcal{L}(x)$ be zero. This gives, using eq.(15), eq.(16) and eq.(14):

$$\begin{aligned} 0 &= \partial_x \mathcal{L}(x) \\ &= \partial_x \left(\frac{1}{N} \sum_i (p_i \cdot x)^2 x^{-2} \right) \\ &= 2 \frac{1}{N} \sum_i \left((p_i \cdot x) p_i x^{-2} - (p_i \cdot x)^2 x^{-3} \right) \\ &= 2 \frac{1}{N} \sum_i (p_i \cdot x) (p_i x - (p_i \cdot x) x^{-1}) \\ &= 2 \frac{1}{N} \sum_i (p_i \cdot x) (p_i \wedge x) x^{-3}. \end{aligned} \quad (21)$$

We multiply by the invertible vector $\frac{1}{2}x^3$, and rewrite the result in the form:

$$\left(\frac{1}{N} \sum_i p_i (p_i \cdot x) \right) \wedge x = 0$$

This outer product equation states that the vector in brackets should be proportional to the vector x . It follows that, to produce an extreme of the $\mathcal{L}$ cost function, x should be an eigenvalue of the operator P defined by the conformal data points $\{p_i\}$:

$$\begin{aligned} P[] : \mathbb{R}^{n+1,1} &\rightarrow \mathbb{R}^{n+1,1} : \\ x &\mapsto \frac{1}{N} \sum_i p_i (p_i \cdot x). \end{aligned} \quad (22)$$

We observe that the cost criterion can be written in terms of this operator as

$$\mathcal{L}(x) = x^{-1} \cdot P[x]. \quad (23)$$

Therefore, denoting an optimal x minimizing $\mathcal{L}(x)$ by x_* and the corresponding eigenvalue of P by λ_* , we have $\mathcal{L}(x_*) = x_*^{-1} \cdot P[x_*] = \lambda_* x_*^{-1} \cdot x_* = \lambda_*$. We will find later (in Lemma 3) that P has a negative eigenvalue corresponding to an imaginary eigensphere. To minimize the cost of $\mathcal{L}$ with a real sphere, we should choose λ_* to be the minimal non-negative value. We summarize:

The sphere $x_* = c - \frac{1}{2}\rho^2 n_\infty$ minimizing the mean of approximate squared distances of a set of conformal points $\{p_i\}$ is the normalized eigenvector of minimum non-negative eigenvalue of the linear operator $P[] = \frac{1}{N} \sum_i p_i (p_i \cdot [])$.

This reduction of hypersphere fitting to an eigenproblem lends itself readily to implementation. The sphere parameters can be extracted from x_* after normalization (i.e., dividing by $-n_\infty \cdot x_*$). The center $\mathbf{c}$ is the Euclidean part of the normalized x_* , and $\rho = \sqrt{x_*^2}$.

Our formulation of the sphere fitting problem in CGA is not new. We did it ourselves in [8] but were

not able to solve the minimization problem properly, instead imposing the denominator as a constraint $x^2 = 1$ on the solution. Then [20] solved it more properly by imposing the constraint by a Lagrange multiplier, but they resorted to a conversion to matrix algebra form and to a solution by constrained optimization; also, they believed they were minimizing a plausibly rescaled *tangential* distance of points to spheres, rather than a good approximation to the radial distance. Neither of these CGA approaches to the hypersphere fitting problem realized the deep connection to hypercircle and k -sphere fitting, which would truly have employed the benefits of the structural CGA representation.

3.3 Implementation

For the implementation of the solution, we can employ a standard linear algebra package to solve the eigenproblem. No geometric algebra facilities are required, the solution can be expressed completely in terms of linear algebra. We only need to remember that the dot product within the operator $P[\cdot]$ should involve the indefinite metric matrix $[M]$ of CGA.

1. Represent a conformal vector $p = p_o n_o + \mathbf{p} + p_\infty n_\infty$ on the basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n, n_o, n_\infty\}$ (where the $\{\mathbf{e}_i\}$ form an orthonormal basis for $\mathbb{R}^n$) as the column matrix $[p] = [\mathbf{p}^T, p_o, p_\infty]^T$. The matrix form $[P]$ of the operator P is then on the same basis:

$$\begin{aligned} [P] &= \frac{1}{N} \Sigma_i [p_i] [p_i]^T [M] \\ &= \frac{1}{N} \Sigma_i [\mathbf{p}_i^T, 1, \frac{1}{2} \|\mathbf{p}_i\|^2] \begin{bmatrix} \mathbf{p}_i \\ 1 \\ \frac{1}{2} \|\mathbf{p}_i\|^2 \end{bmatrix} \begin{bmatrix} [1] & \mathbf{0} & \mathbf{0} \\ \mathbf{0}^T & 0 & -1 \\ \mathbf{0}^T & -1 & 0 \end{bmatrix} \\ &= \frac{1}{N} \Sigma_i \begin{bmatrix} \mathbf{p}_i \mathbf{p}_i^T & -\frac{1}{2} \|\mathbf{p}_i\|^2 \mathbf{p}_i & -\mathbf{p}_i \\ \mathbf{p}_i^T & -\frac{1}{2} \|\mathbf{p}_i\|^2 & -1 \\ \frac{1}{2} \|\mathbf{p}_i\|^2 \mathbf{p}_i^T & -\frac{1}{4} \|\mathbf{p}_i\|^4 & -\frac{1}{2} \|\mathbf{p}_i\|^2 \end{bmatrix} \quad (24) \end{aligned}$$

where $[1]$ is the $n \times n$ identity matrix. As usual in such problems, one can construct this matrix efficiently by setting up a data matrix $[D]$ with column i equal to $[p_i]$, and forming $[P] = [D] [D]^T [M] / N$.

2. Solve the eigenproblem for this $[P]$, selecting the minimum non-negative eigenvalue λ_* and the corresponding eigenvector x_* .
3. The actually achieved minimum cost $\mathcal{L}(x_*)$ of the fitted hypersphere equals λ_* . To interpret the hypersphere that achieves it, normalize the eigenvector $[x_*]$ to have its n_o -component $[x_*]_o$ be equal to 1. It is then the representation of the best-fit hypersphere as $[\mathbf{c}^T, 1, \frac{1}{2}(\|\mathbf{c}\|^2 - \rho^2)]^T$. Therefore the

hypersphere's center $\mathbf{c}$ is found as the first n components of the normalized $[x_*]$, and the radius as $\rho = \sqrt{\|\mathbf{c}\|^2 - 2[x_*]_\infty}$.

Figure 4 shows a simulation two spheres fitted to a set of noisy points by this method, compared to the ground truth that generated the data.

The complexity of this algorithm in n dimensions with N data points is $O(N(n+2))$ for the construction of the matrix $[P]$ plus at most $O((n+2)^3)$ for the eigendecomposition.

Though the geometric method of construction and solution is different, this fit is effectively the same as the algebraic fit by Pratt [18]. We will now explain this and compare to some other proposed fits.

4 Comparison with Other Methods

As mentioned in the introduction, the various circle fits studied in [1] can easily be generalized to hypersphere fits. The results from [1] then apply: for reasonable noise and less than a thousand data points, they are all optimal in achieving the KCR lower bound for their MSE. Our cost criterion for hyperspheres is the n -D generalization of the circle fit by Pratt [18], also treated in [1]. It was chosen as our representative fit of choice, because of its algebraic properties that will give us the fits for k -spheres in n -D in this paper. Also, as we will discover in Section 7, it is a template for the hyperaccurate fit of hyperspheres.

4.1 Equivalence to the Pratt fit

In [1], the Pratt fit is characterized as an ‘algebraic fit’ having the following constrained cost function for points with coordinates $\mathbf{p}_i = (x_i, y_i)$ and circle characterization parameters (A, B, C, D) :

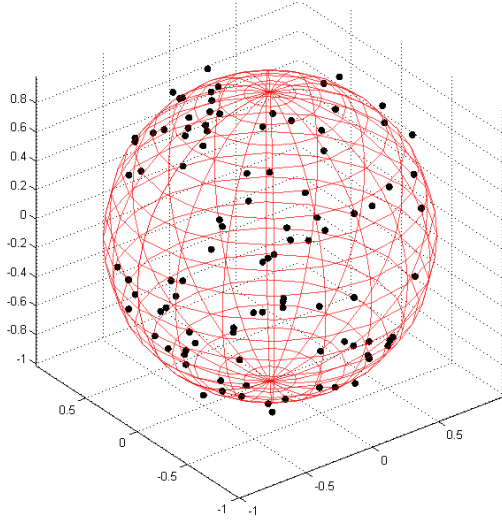
$$\begin{aligned} \text{Minimize } & \Sigma_i (A(x_i + y_i)^2 + Bx_i + Cy_i + D)^2 \\ \text{while } & B^2 + C^2 - 4AD = 1. \end{aligned}$$

Rewrite this to:

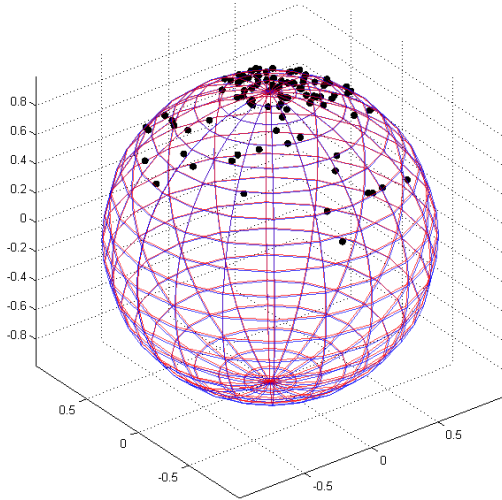
$$\begin{aligned} \text{Minimize } & 4A^2 \Sigma_i \left(\left(x_i + \frac{B}{2A} \right)^2 + \left(y_i + \frac{C}{2A} \right)^2 - \frac{B^2 + C^2 - 4AD}{4A^2} \right)^2 \\ \text{while } & B^2 + C^2 - 4AD = 1. \end{aligned}$$

We recognize this as a parametrization of the squared distance of the point p_i to the circle center $\mathbf{c} = (-B, -C)/2A$, relative to the squared radius $\rho^2 = \frac{B^2 + C^2 - 4AD}{4A^2}$, which was indeed Pratt’s original motivation:

$$\text{Minimize } \Sigma_i 4A^2 (\|\mathbf{p}_i - \mathbf{c}\|^2 - \rho^2)^2 \text{ while } 4A^2 \rho^2 = 1.$$



(a)



(b)

Fig. 4 3D sphere fitting using our eigenmethod implemented in Matlab. Two noisy sets of 100 data points were generated in simulation from the blue spheres. Red denotes the fitted sphere. In (a), the point data distribution is uniform over the unit sphere with a radial standard deviation of 0.05 times the radius; the original and fit are indistinguishable. In (b) the data has a limited solid angle spread of 0.5 radian standard deviation, with radial standard deviation of 0.01 times the radius.

Assuming $A \neq 0$, this has the same solution as the optimization problem:

$$\text{Minimize } \sum_i \frac{(\|\mathbf{p}_i - \mathbf{c}\|^2 - \rho^2)^2}{\rho^2}.$$

In CGA, with a circle parametrized as $x = c - \frac{1}{2}\rho^2 n_\infty$ where c is the conformal point at the center, we have $\|\mathbf{p}_i - \mathbf{c}\|^2 - \rho^2 = -2\mathbf{p}_i \cdot x$, and $\rho^2 = x^2$. Therefore this can be written as:

$$\text{Minimize } 4 \sum_i \frac{(p_i \cdot x)^2}{x^2},$$

which is the formulation of our fit (with an irrelevant factor of $4N$), motivated in Section 3.1 by geometrical reasoning on an approximation of the squared distance. This final form in CGA of the Pratt fit will lend itself well to addressing the k -sphere fitting.

4.2 Other algebraic methods

Exact minimization of eq.(1) is expensive (a typical Levenberg-Marquardt iteration takes 20 times as long as the eigenproblem for P ; and depending on the quality of the seed, several iterations are required for a good fit). Therefore people have long studied ways of constraining the algebraic sphere equation parameters to cheaply find a stable solution to the fitting problem by linear algebra. The solutions given are not always geometrically meaningful: some of the early algebraic methods were not even translation invariant! Unfortunately, the need to provide stable numerical methods is frequently founded in trusted measures like the Frobenius norm between data matrices [4], [12], which is not a good indicator of the Euclidean distance between data points and fitted sphere. For instance, to perform a ‘seamless’ transition to hyperplane fits, Nievergelt [16] concentrates on the algebraic representation of spheres and planes to produce a constrained total least squares solution (CTLTS) - but the ‘total least squares’ is in the sense of a Frobenius norm on projected data matrices, rather than in terms of a geometrical distance measure on the point data.

While such methods may perform well enough when fitting a sphere to point data that is widely enough distributed over the intended surface, they tend to fail quickly when the data is more limited, since their optimization measures cannot capture the Euclidean ‘sphericity’ of the data. For the fits of [4], [12] and [16], this is illustrated in Figure 1, and more extensively in the simulations of Appendix A.

In this paper, we achieve the connection to powerful standard linear algebra techniques in a subtly different way: by the isometric CGA embedding in a higher

dimensional space, where the dot product directly indicates the Euclidean distance. In this representation, what is sensible algebraically is sensible geometrically. Some of the earlier algebraic techniques can be reformulated in this framework, showing their essentially geometric motivation more clearly - we have just done this for Pratt's fit, and will show the hyperaccurate fit in Section 7. Not surprisingly, such fits were found to be among the best algebraic methods available in the analysis by [1] (see also Appendix A).

Conceptually, a new feature of our CGA-based solution to hypersphere fitting is the explicit form as a pure eigenproblem of a linear operator, namely P of eq.(22). This contrasts with earlier solutions, which are either algorithmic (when based on the least square criterion) or phrased in terms of a generalized eigenproblem of matrix operators such as in [1]'s treatment of the algebraic fits. As we will see below, this operator P is like an inertia matrix in the conformal space, and the hypersphere fit therefore becomes algebraically very analogous to finding the major axis of a distribution by means of its covariance matrix; where one should use the distribution of conformal point vectors from the conformal space with its indefinite metric, rather than of Euclidean direction vectors.

In summary, as an 'algebra of spheres', CGA is a metrically meaningful uniform representation of the three types of elements (hyperspheres, hyperplanes and data points), and thus incidentally achieves Nievergelt's goal of 'seamlessness'. The only subtlety we have to be aware of is that the indefinite metric produces results that are unfamiliar to our usual (implicit) assumption of a Euclidean metric when applying our linear algebra techniques. In what follows, we will therefore carefully develop our analysis of what is essentially the eigenstructure of a symmetric operator in an indefinite metric, with as secondary purpose a demonstration of the power of CGA's ability to perform this analysis directly in terms of the geometric elements of the problem.

5 Hypercircle Fitting

We define a *hypercircle* as the intersection of a hypersphere and a hyperplane. Fitting hypercircles to a set of points in an approximate least square manner is a delightfully minor extension of the hypersphere approach. It uses the ability of any geometric algebra to process 2-blades as easily as vectors.

5.1 An Approximate Least Squares Criterion

Intersecting a hypersphere s and a hyperplane π passing through its center produces a great circle on that sphere. In CGA, this great circle X is (dually) represented by $X = s \wedge \pi$, see eq.(12). The condition that π passes through the center of s is geometrically equivalent to s and π intersecting orthogonally; this is represented in CGA by $s \cdot \pi = 0$.

We now simply substitute X for x in the distance measure $(p \cdot x)^2/x^2$, and demonstrate that this provides a sensible approximate squared distance of the point p to the circle X . But since we have to make sure that the result is a scalar, also in our later extension to k -spheres, we replace the squares by scalar products. We first note that

$$\begin{aligned} X * X &= (\pi \wedge s) * (\pi \wedge s) \\ &= \pi * (s \cdot (\pi \wedge s)) \\ &= \pi * ((s \cdot \pi) s - \pi (s \cdot s)) \\ &= -(\pi \cdot \pi) (s \cdot s). \end{aligned}$$

Expanding the inner product of p with X in terms of the factors of X gives:

$$p \cdot X = p \cdot (s \wedge \pi) = (p \cdot s) \pi - (p \cdot \pi) s,$$

and its scalar product square simplifies due to $s \cdot \pi = 0$:

$$(p \cdot X) * (p \cdot X) = (p \cdot s)^2 \pi \cdot \pi + (p \cdot \pi)^2 s \cdot s + 0.$$

It follows that

$$-\frac{(p \cdot X) * (p \cdot X)}{X * X} = (p \cdot s)^2/s^2 + (p \cdot \pi)^2/\pi^2. \quad (25)$$

The first term we recognize as 'the approximate squared distance from the point p to the hypersphere'. The second term is 'the exact squared distance from the point p to the carrier hyperplane'. For using the hyperplane representation eq.(8), we find for a point p :

$$\begin{aligned} \frac{(p \cdot \pi)^2}{\pi^2} &= \frac{(p \cdot (\mathbf{n} + (q \cdot \mathbf{n})n_\infty))^2}{\mathbf{n}^2} \\ &= \frac{((\mathbf{p} - \mathbf{q}) \cdot \mathbf{n})^2}{\mathbf{n}^2} \\ &= ((\mathbf{p} - \mathbf{q}) \cdot \hat{\mathbf{n}})^2, \end{aligned} \quad (26)$$

which is the square of the distance of a point at location $\mathbf{p}$ relative to a plane with unit normal vector $\hat{\mathbf{n}}$ passing through location $\mathbf{q}$.

Together, these terms of eq.(25) therefore provide a sensible local squared distance to the hypercircle X , by means of a local application of Pythagoras' theorem to these squared distances to locally perpendicular elements. (Note the minus sign we had to include in

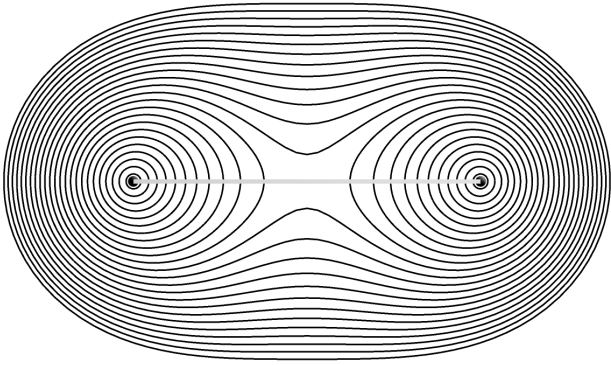


Fig. 5 Contour lines of the function $\sqrt{-(p \cdot X) * (p \cdot X) / (X * X)}$ occurring in the cost function eq.(27), as a function of the location p , for a 3D circle X seen in cross section through a diameter as the two black dots. (Fig. 6.9 in [17], reproduced by kind permission of its author.)

eq.(25); in Section 6 we will reformulate more generally to make it appear automatically.)

The sum of such measures for each point p_i then approximates the sum of squared distances to the hypercircle X . That sum tends to cancel the approximation errors of points inside and outside the encompassing sphere, and is a second order approximation to the exact measure when the points are not too asymmetrically distributed around the best fit hypercircle.

A plot of the contour lines of the distance function $\sqrt{-(p \cdot X) * (p \cdot X) / (X * X)}$ occurring in the cost function eq.(27), as a function of the location of p , is given in Figure 5, for a cross-section of a 3D circle X (represented as a 2-blade in CGA) through a plane containing its center. For points close to the circle, the contour lines are almost circular, with the deviation from circularity inside and outside the circle apparently canceling in first order.

5.2 Optimal Hypercircle Fitting as Eigenproblem

We need to solve, for a bivector X :

Find X that minimizes

$$\mathcal{L}(X) = -\frac{1}{N} \sum_i \frac{(p_i \cdot X) * (p_i \cdot X)}{X * X}. \quad (27)$$

Differentiation to the bivector X requires some specific results from multivector differentiation, notably eq.(15), eq.(17) and eq.(16), made applicable by rewrit-

ing using eq.(9):

$$\begin{aligned} 0 &= \partial_X \mathcal{L}(X) \\ &= -\frac{1}{N} \sum_i \left(\partial_X ((p_i \cdot X) * (p_i \cdot X)) (X * X)^{-1} \right. \\ &\quad \left. + ((p_i \cdot X) * (p_i \cdot X)) \partial_X (X * X)^{-1} \right) \\ &= -\frac{1}{N} \sum_i \left(\partial_X (((p_i \cdot X) \wedge p_i) * X) (X * X)^{-1} \right. \\ &\quad \left. - 2((p_i \cdot X) * (p_i \cdot X)) X (X * X)^{-2} \right) \\ &= -\frac{1}{N} \sum_i \left(2(p_i \cdot X) \wedge p_i (X * X)^{-1} \right. \\ &\quad \left. + 2\mathcal{L}(X) X (X * X)^{-1} \right) \\ &= \frac{2}{X * X} \left(\frac{1}{N} \sum_i p_i \wedge (p_i \cdot X) - \mathcal{L}(X) X \right). \end{aligned}$$

We thus obtain the equation that X needs to satisfy to be an extreme solution X_* :

$$\frac{1}{N} \sum_i p_i \wedge (p_i \cdot X_*) = \mathcal{L}(X_*) X_*. \quad (28)$$

Therefore X_* is the eigenbivector for smallest non-negative eigenvalue $\mathcal{L}_* = \mathcal{L}(X_*)$ of the operator on the left:

$$\begin{aligned} P' : \bigwedge^2 \mathbb{R}^{n+1,1} &\rightarrow \bigwedge^2 \mathbb{R}^{n+1,1} : \\ X &\mapsto \frac{1}{N} \sum_i p_i \wedge (p_i \cdot X). \end{aligned} \quad (29)$$

This operator has the form of the CGA equivalent of an inertia tensor, a bivector-valued linear mapping on bivectors occurring in classical dynamics for Euclidean bivectors, see [6] – but here, the inertia is in the Minkowski space $\mathbb{R}^{n+1,1}$.

5.3 Eigenstructure of the Conformal Inertia Tensor P'

The solutions to the optimization problem are eigenbivectors of P' . Since we want to fit circles, we are only interested in 2-blades among the solutions. We could have imposed the 2-blade condition $X \wedge X = 0$ of eq.(13) explicitly by means of a Lagrange multiplier on the cost criterion. However, as we now show, this condition is actually superfluous, because *all* extrema of the fitting problem are blades (at least if the eigenvalues of P' are all different).

The proof exploits the very similar form of P' acting on bivectors (above) to P acting on vectors of eq.(22). We can expose this more explicitly by rewriting P slightly, replacing the product with a scalar by an equivalent outer product.

$$P : \mathbb{R}^{n+1,1} \rightarrow \mathbb{R}^{n+1,1} : x \mapsto \sum_i p_i \wedge (p_i \cdot x). \quad (30)$$

This shows that P and P' only differ in the arguments they are applied to.⁵ We may therefore denote both by

⁵ Those familiar with GA should note that P' is *not* simply an outermorphism extension of P , see eq.(34).

P , but for clarity in the following discussion it is more pertinent to retain their individual notations a while longer.

5.3.1 Why All Bivector Extrema Are 2-Blades

First, we relate the eigenbivectors of P' to the eigenvectors of P .

Lemma 1 *Let x_j and x_k be eigenvectors of P , with eigenvalues λ_j and λ_k , respectively. Then $x_j \wedge x_k$ is an eigen-2-blade of P' with eigenvalue $(\lambda_j + \lambda_k)$.*

Proof By the assumption, x_j satisfies: $P[x_j] = \sum_i p_i \wedge (p_i \cdot x_j) = \lambda_j x_j$, and so does x_k . Now evaluate $P'[x_j \wedge x_k]$, using the dot product expansion rule eq.(10):

$$\begin{aligned} P'[x_j \wedge x_k] &= \sum_i p_i \wedge \left(p_i \cdot (x_j \wedge x_k) \right) \\ &= \sum_i p_i \wedge \left((p_i \cdot x_j) x_k - x_j (p_i \cdot x_k) \right) \\ &= P[x_j] \wedge x_k - P[x_k] \wedge x_j \\ &= \lambda_j x_j \wedge x_k - \lambda_k x_k \wedge x_j \\ &= (\lambda_j + \lambda_k) x_j \wedge x_k. \quad \square \end{aligned}$$

Lemma 2 *If all eigenvalues of P' are different, all eigenbivectors of P' are 2-blades.*

Proof The operator P has an eigenbasis of $(n+2)$ vectors. From these we can construct $\binom{n+2}{2}$ elements that are 2-blades. Each of those is an eigenbivector (even an eigen-2-blade) of P' , by lemma 1. P' determines an eigenequation for bivectors, and so if its eigenvalues are all different, P' should have $\binom{n+2}{2}$ linearly independent ‘eigenvectors’ in this bivector space. It follows that we can construct each eigenbivector of P' from eigenvectors of P . Therefore all eigenbivectors of P' are 2-blades. $\square$

If some of the eigenvalues of P' were the same, a linear combination of 2-blades with the same eigenvalue could be combined into an eigenbivector that is not necessarily a blade (for instance, if $\lambda_1 + \lambda_4 = \lambda_2 + \lambda_3 \equiv \lambda$, then $x_1 \wedge x_4 + x_2 \wedge x_3$ is an eigenbivector of P' , with eigenvalue λ , but not necessarily an eigenblade). With an ordering of eigenvalues $x_1 \leq x_2 \leq x_3 \leq x_4$, this implies that all four eigenvalues should be the same.⁶ Nevertheless, we can always construct the eigencircles for P' as intersections of the eigenspheres of P . That P' may have other extrema than these eigencircles need not concern us in our application.

⁶ The simplest situation in which this occurs (barring a trivial P based on fully coincident data points) is two data points in $\mathbb{R}^4$, which gives four zero eigenvalues to exactly fit the dual point pair (0-sphere) $x_1 \wedge x_2 \wedge x_3 \wedge x_4$.

5.3.2 An Orthonormal Eigenbasis of Hyperspheres; Nature of the Extrema

The manner in which we proved the result on the superfluity of the 2-blade condition suggests an alternative manner of computing the optimal circle: as the intersection of the two best fit hyperspheres. To understand this geometrically, we investigate the geometry of the eigenvectors of P , and the ordering of their eigenvalues.

Since P of eq.(22) is a symmetric linear mapping, its eigenvectors are orthogonal (more precisely, for different eigenvalues, they are, and for multiple eigenvalues, they can be made to be). Therefore the $(n+2)$ hyperspheres that are the eigenvectors of P can be made into an orthonormal basis of hyperspheres (normalizing a hypersphere s is done through dividing by the scalar $-n_\infty \cdot s$). Geometrically, these are hyperspheres that intersect at right angles (if both are real; if one is real and the other imaginary, they intersect in a great circle of the imaginary sphere, see [8]).

To prevent a totally degenerate P with only zero eigenvalues, we assume in the following that the number of non-coincident data points is at least 2.

Lemma 3 *P has $n+1$ real eigenhyperspheres corresponding to non-negative eigenvalues, and one imaginary eigenhypersphere corresponding to a negative eigenvalue.*

Proof The eigenvalue of a solution x is the value of $\mathcal{L}(x)$. The numerator of $\mathcal{L}$, being a sum of squares of scalars, is always non-negative. The denominator is the square of a vector representing a sphere, and is negative when that sphere is imaginary (i.e. when it has a negative squared radius), see Section 2. Thus a negative eigenvalue should correspond to an eigenvector representing an imaginary eigensphere.

We know that the eigenvectors are mutually orthogonal and represent hyperspheres; moreover, the perpendicularity of the vectors is equivalent to the orthogonality of the hyperspheres, see Section 2-Hyperspheres. It was shown in [19] that when $n+2$ hyperspheres are mutually orthogonal in n -space, precisely one is imaginary.

The argument in [19] is that from the above it follows that $\rho_i^2 + \rho_j^2 \geq 0$ for an arbitrary pair of orthogonal hyperspheres with radii ρ_i and ρ_j , so not more than one can have negative square. But [19] also shows that $(n+2)$ mutually orthogonal hyperspheres obey $\sum_{i=1}^{n+2} \rho_i^{-2} = 0$, so that at least one of them must have a negative square. Therefore, precisely one has a negative square.

Incidentally, it is easy to prove the property $\sum_i \rho_i^{-2} = 0$ using CGA, from the orthogonality requirements for

normalized hyperspheres: $n_\infty \cdot x_i = -1$ and $x_i \cdot x_j = x_i^2 \delta_{ij}$ combined with $x_1 x_2 \cdots x_{n+2} = \alpha I_{n+2}$ with I_{n+2} a unit blade of grade $(n+2)$. We find $\Sigma_i x_i^{-1} = -(n_\infty \cdot (x_1 x_2 \cdots x_{n+2})) (x_1 x_2 \cdots x_{n+2})^{-1} = -(n_\infty \cdot I_{n+2}) I_{n+2}^{-1} = -n_\infty$. Then $0 = n_\infty \cdot n_\infty = -n_\infty \cdot \Sigma_i x_i^{-1} = \Sigma_i (-n_\infty \cdot x_i) / x_i^2 = \Sigma_i x_i^{-2} = \Sigma_i \rho_i^{-2}$. $\square$

With this in hand, we can investigate the nature of the extrema of $\mathcal{L}$.

Lemma 4 *If all eigenvalues of P are different and ordered in size $\lambda_0 < 0 \leq \lambda_1 < \cdots < \lambda_{n+1}$, the value of $\mathcal{L}(x_j)$ at the corresponding eigenvalue is: a local maximum at x_0 and x_{n+1} ; a local minimum at x_1 ; a saddle point at the other x_j , with $(j-1)$ orthogonal decreasing directions and $(n+1-j)$ orthogonal increasing directions. Any extremum at x_j has one second order stationary direction x_j .*

Proof We compute the local second order term in the Taylor series of $\mathcal{L}(x_j)$ around an eigenhypersphere x_j , by perturbing the derivative $\partial_x \mathcal{L}(x) = 2(P[x] \wedge x) x^{-3}$. We use the fact that x_j is an eigenvector of P :

$$\begin{aligned} \mathcal{L}(x_j + \epsilon) - \mathcal{L}(x_j) &= \frac{1}{2} \epsilon \cdot \left(\partial_x \mathcal{L}(x_j + \epsilon) - \partial_x \mathcal{L}(x_j) \right) + O(\epsilon^3) \\ &\approx \epsilon \cdot \left((P[\epsilon] \wedge x_j + P[x_j] \wedge \epsilon) x_j^{-3} \right. \\ &\quad \left. + (P[x_j] \wedge x_j) ((x_j + \epsilon)^{-3} - x_j^{-3}) \right) \\ &= \epsilon \cdot \left((P[\epsilon] \wedge x_j + \lambda x_j \wedge \epsilon) x_j^{-3} \right) + 0 \\ &= \epsilon \cdot \left(((P[\epsilon] - \lambda \epsilon) \wedge x_j) x_j^{-3} \right). \end{aligned}$$

Now consider the arbitrary change ϵ , and decompose it onto the orthonormal eigenbasis of P . Take a change in ϵ_i the i -th eigendirection at an extremum determined by the j -th eigenvector x_j , and call the corresponding second order term of the Taylor series $\Delta(\epsilon_i, x_j)$. Then we obtain:

$$\begin{aligned} \Delta(\epsilon_i, x_j) &= \epsilon_i \cdot \left(((P[\epsilon_i] - \lambda_j \epsilon_i) \wedge x_j) x_j^{-3} \right) \\ &= \epsilon_i \cdot \left(((\lambda_i - \lambda_j) \epsilon_i) \wedge x_j \right) x_j^{-3} \\ &= (\lambda_i - \lambda_j) \epsilon_i^2 / x_j^2, \end{aligned}$$

since when the eigendirections i and j are the same the result is zero, and otherwise the eigenvectors ϵ_i and x_j are orthogonal, simplifying the products. It is immediately clear that each extremum x_j has a stationary direction ϵ_j .

Consider first the extremum at x_0 , with $x_0^2 < 0$. There $\lambda_0 = -|\lambda_0|$, all other $\lambda_i \geq 0$, all $\epsilon_i^2 > 0$ for $i \neq 0$, and so $\Delta(\epsilon_i, x_0) < 0$ for $i \neq 0$. Therefore there is a local maximum at $\mathcal{L}(x_0)$.

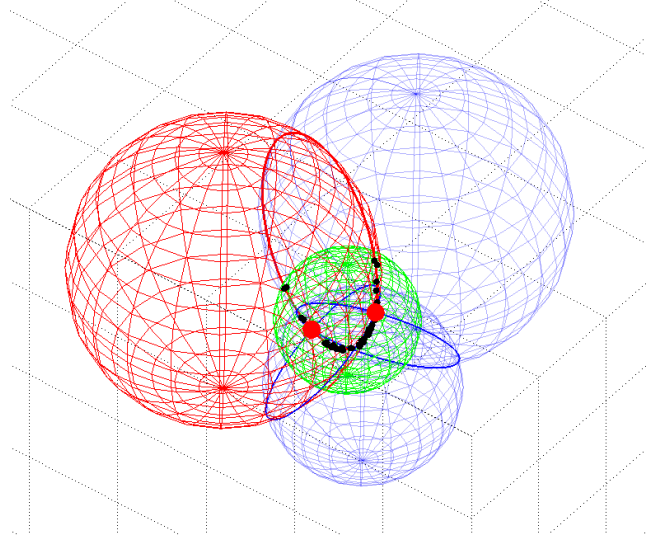


Fig. 6 A set of 50 data noisy points (in black) was generated from a 3D circle in simulation. From the set of data points, the conformal inertia matrix P was constructed. Its eigenvectors represent a set of 5 orthogonal spheres. We show the first 3 real spheres (in blue and red, intersecting perpendicularly) and the imaginary eigensphere (in green, intersected by the others in an equator). The best fit sphere (in red) is the intersection of the two best fitting real hyperspheres, i.e. the ones with lowest nonnegative eigenvalues. The best fit point pair (in red) is the intersection of the best fit circle with the next best fit sphere.

Now consider some x_j with $j \neq 0$, and ϵ_i for $0 < i < j$. Since $\lambda_i < \lambda_j$ while $\epsilon_i^2 > 0$ and $x_j^2 > 0$, we get $\Delta(\epsilon_i, x_j) < 0$ for $0 < i < j$, implying $(j-1)$ decreasing directions. Similarly, we obtain $\Delta(\epsilon_i, x_j) > 0$ for $j < i \leq n+1$, implying $(n+1-j)$ increasing directions. For $i = 0$, we obtain $\Delta(\epsilon_0, x_j) > 0$, so this is an increasing direction for all $x_j \neq x_0$. The lemma now follows. $\square$

The corollary for real hyperspheres is:

Corollary 1 *If the eigenvalues of P are all different, the unique non-negative minimum for $\mathcal{L}(x)$ is at the eigenvector of P of smallest non-negative eigenvalue.*

One can trivially refine the lemmas when multiple eigenvalues occur, but since this is unlikely to happen in practice with a P based on a realistic amount of real data for the fit, we will not get into that administrative exercise.

With these results, we can determine the optimal hypercircle to fit the data p_i , by finding the two hyperspheres with minimum nonnegative eigenvalues of P , and then computing the optimal hypercircle as their intersection. This is illustrated in Figure 6. The principle will generalize to the fitting of k -spheres in Section 6.

5.4 Implementation of Circle Fitting in 3D

As we have shown, the bladeness of the eigensolutions of P' is guaranteed if the eigenvalues of P are all different; and in practical situations based on real point data, this is likely to be the case. We present two ways of implementing the eigenproblem for P' : directly by constructing its matrix $[P']$, and indirectly from the eigenvectors of the matrix $[P]$. We show how to convert these solutions into matrix language for the specific case of 3D. Both of course produce the same result.

5.4.1 Direct solution of $[P']$

The matrix $[P']$ of the linear map P' can be written as a sum of matrices $[P'_i]$ contributed by each data point p_i . Since P' is a linear map from bivectors to bivectors, we need to introduce a bivector basis to represent it as a matrix. This requires $\binom{n+2}{2}$ elements in general. For $n = 3$, we use the 10-D bivector basis $\{e_{23}, e_{31}, e_{12} \mid e_{o1}, e_{o2}, e_{o3} \mid e_{1\infty}, e_{2\infty}, e_{3\infty} \mid e_{o\infty}\}$. Then $[P']$ can be evaluated by applying it to the basis bivectors and tabulating the resulting components. The result is:

$$[P'] = \frac{1}{N} \sum_i \begin{bmatrix} -[\mathbf{p}_i^\times]^2 & -\frac{1}{2}\mathbf{p}_i^2 [\mathbf{p}_i^\times] & [\mathbf{p}_i^\times] & 0 \\ -[\mathbf{p}_i^\times] & [\mathbf{p}_i^\times]^2 + \frac{1}{2}\mathbf{p}_i^2 [1] & [1] & -[\mathbf{p}_i] \\ [\mathbf{p}_i^\times] & \frac{1}{4}\mathbf{p}_i^4 [1] & [\mathbf{p}_i^\times]^2 + \frac{1}{2}\mathbf{p}_i^2 [1] & -\frac{1}{2}[\mathbf{p}_i^3] \\ 0^T & \frac{1}{2}[\mathbf{p}_i^3]^T & [\mathbf{p}_i]^T & -\mathbf{p}_i^2 \end{bmatrix}$$

where $[1]$ denotes the 3×3 identity matrix, $[\mathbf{p}_i^\times]$ is the cross product matrix, $\mathbf{p}_i^2$ equals the more classical $\|\mathbf{p}_i\|^2$, and $[\mathbf{p}_i^3] = \mathbf{p}_i^2 [\mathbf{p}_i]$. One computes the eigenvector of smallest nonnegative eigenvalue of this matrix using a linear algebra package.

To unpack the resulting 10D eigen(bi)vector into meaningful circle parameters, we use the general expression for a dual circle κ in CGA, as the intersection of a hyperplane with normal vector $\mathbf{n}$ containing c , and a sphere with radius ρ around c :

$$\begin{aligned} \kappa &= \alpha (\mathbf{n} + (\mathbf{c} \cdot \mathbf{n}) n_\infty) \wedge (n_o + \mathbf{c} + \frac{1}{2}(\mathbf{c}^2 - \rho^2) n_\infty) \\ &= \alpha (\mathbf{n} \wedge n_o + \mathbf{n} \wedge \mathbf{c} - (\mathbf{c} \cdot \mathbf{n}) n_o \wedge n_\infty \\ &\quad + (\frac{1}{2}(\mathbf{c}^2 - \rho^2) \mathbf{n} - (\mathbf{c} \cdot \mathbf{n}) \mathbf{c}) n_\infty). \end{aligned} \quad (31)$$

- Thus $-\alpha \mathbf{n}$ can be retrieved immediately as the components of $\{e_{oi}\}$, normalization of $\mathbf{n}$ then splits it in α and $\mathbf{n}$ if necessary.
- The components of the Euclidean and $e_{o\infty}$ parts then give the outer and inner product of $\mathbf{c}$ and $\mathbf{n}$ (and therefore effectively their invertible geometric

product by eq.(14)), from which $\mathbf{c}$ can be recovered. To show how, suppose that $\mathbf{n} \wedge \mathbf{c}$ has the bivector coefficients $[B_1, B_2, B_3]^T$ on the basis $\{e_{23}, e_{31}, e_{13}\}$, and that the value of $\mathbf{n} \cdot \mathbf{c}$ equals B_0 . The 3D cross product is embedded in geometric algebra as the dual of a 2-blade, see [8]. The coefficients of the cross product $\mathbf{n} \times \mathbf{c}$ on the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ turn out to be the same as the 2-blade coefficients $[B_1, B_2, B_3]^T$ on the basis $\{e_{23}, e_{31}, e_{13}\}$. Therefore, given $\mathbf{n}$ and the B_i , solving the matrix equation

$$\begin{bmatrix} n_1 & n_2 & n_3 \\ 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{bmatrix} [\mathbf{c}] = \begin{bmatrix} B_0 \\ B_1 \\ B_2 \\ B_3 \end{bmatrix} \quad (32)$$

yields $\mathbf{c}$. Since the matrix is of rank 3 the solution is well-defined, and explicitly given by

$$[\mathbf{c}] = \begin{bmatrix} B_0 & -B_3 & B_2 \\ B_3 & B_0 & -B_1 \\ -B_2 & B_1 & B_0 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} / (n_1^2 + n_2^2 + n_3^2) \quad (33)$$

Effectively, this is how division under the geometric product in 3D is implemented in linear algebra.

- With α , $\mathbf{n}$ and $\mathbf{c}$ known, ρ^2 can be derived from the $\{e_{i\infty}\}$ component vector $\mathbf{v}_\infty$ as $\rho^2 = \|\mathbf{c}\|^2 - 2\mathbf{n} \cdot \mathbf{v}_\infty / \alpha - 2(\mathbf{c} \cdot \mathbf{n})^2$.

Figure 7 gives typical results of this fit.

5.4.2 Indirect solution from $[P]$

The geometrical insight from Section 5.3.1 suggests another implementation: we do not need to construct the $\binom{n+2}{2} \times \binom{n+2}{2}$ -dimensional matrix $[P']$ and solve its eigenproblem, but we can limit ourselves to the $(n+2) \times (n+2)$ matrix $[P]$ of eq.(24). The intersection of its two best eigenspheres give the best fit circle.

In this approach, we therefore have to implement the outer product of CGA, to construct the 2-blade solution from its two best eigenvectors. This is fairly straightforward: an outer product $y \wedge x$ can be considered as a linear operator on x , parametrized by y . In 3D CGA, choosing the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, n_o, n_\infty\}$ for x and y , with correspondingly labeled coordinates, and as bivector basis $\{e_{23}, e_{31}, e_{12} \mid e_{o1}, e_{o2}, e_{o3} \mid e_{1\infty}, e_{2\infty}, e_{3\infty} \mid e_{o\infty}\}$, we have by straightforward computation of the components of the outer product:

$$[y \wedge x] = \begin{bmatrix} [\mathbf{y}^\times] & \mathbf{0} & \mathbf{0} \\ y_o [1] & -\mathbf{y} & 0 \\ -y_\infty [1] & 0 & \mathbf{y} \\ \mathbf{0}^T & -y_\infty y_o & \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ x_o \\ x_\infty \end{bmatrix}.$$

Interpretation of the resulting 2-blade in terms of classical circle parameters is done as above, based on eq.(31).

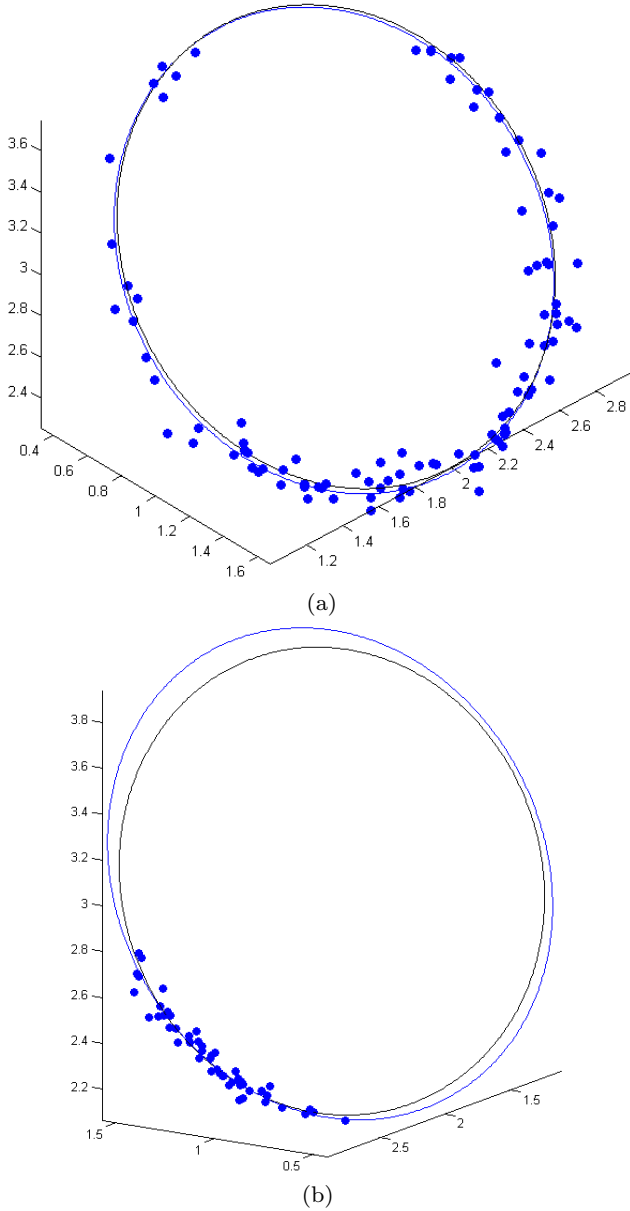


Fig. 7 3D circle fitting using our eigenmethod implemented in Matlab. Two noisy sets of 100 and 50 data points generated in simulation from the blue circle, and in black the fitted circles. In (a), the point data distribution is fairly uniform over the unit circle with a radial standard deviation of 0.01 radius, in (b) the data has a limited solid angle spread of 0.5 radian standard deviation, with radial standard deviation of 0.005 radius.

Figure 6 illustrates this method of obtaining the best fit 3D circle.

6 Fitting General Rounds S^k in n -D Euclidean Space.

The procedure we have used can be extended to fit general ‘rounds’, i.e., k -dimensional spherical surfaces S^k

to point data in an n -dimensional Euclidean space. We may consider such rounds as the intersection of a hypersphere with orthogonal hyperplanes through its center. In CGA, that construction is (dually) represented as the outer product of the orthogonal vectors representing the hypersphere and the hyperplanes: $X = s \wedge \pi_1 \wedge \dots \wedge \pi_{\ell-1}$, which produces an ℓ -blade. In n -D, using 1 term represents the spherical surface mathematicians call S^{n-1} , the $(n-1)$ -sphere; using 2 terms gives S^{n-2} etc., so that we need to set $\ell = n - k$ to get a k -sphere S^k in n -D Euclidean space represented as an ℓ -blade.

6.1 Defining the Cost Criterion

We now generalize our fit of spheres and circles to k -spheres, by extending our formulas to ℓ -vectors. We introduce the following grade-preserving function on ℓ -vectors in CGA:

$$P : \bigwedge^\ell \mathbb{R}^{n+1,1} \rightarrow \bigwedge^\ell \mathbb{R}^{n+1,1} :$$

$$X \mapsto P[X] = \frac{1}{N} \sum_{i=1}^N p_i \wedge (p_i \cdot X).$$

This operator P has a useful property (straightforward to prove from eq.(10) and the antisymmetry of the outer product):

$$P[x \wedge X] = P[x] \wedge X + x \wedge P[X]. \quad (34)$$

and it is symmetric (a consequence of eq.(9)):

$$Y * P[X] = P[Y] * X.$$

We define our general cost criterion as:

$$\mathcal{L}(X) = \frac{X * P[X]}{X * X}.$$

To show that this represents a reasonable criterion, we expose its recursive structure. We compute the cost function of a blade $x \wedge X$ with a vector factor x and a general blade X , mutually orthogonal (so that $x \cdot X = 0$). It can be written as a sum of cost functions of its

orthogonal parts:

$$\begin{aligned}
& \mathcal{L}(x \wedge X) \\
&= \frac{(x \wedge X) * P[x \wedge X]}{(x \wedge X) * (x \wedge X)} \\
&= \frac{(X \wedge x) * P[x \wedge X]}{(X \wedge x) * (x \wedge X)} \\
&= \frac{(X \wedge x) * (P[x] \wedge X + x \wedge P[X])}{(X \wedge x) * (x \wedge X)} \\
&= \frac{(x * P[x]) (X * X) + (x * x) (X * P[X])}{(x * x) (X * X)} \\
&\quad - \frac{X * (x \wedge (x * P[X]))}{(x * x) (X * X)} \\
&= \frac{x * P[x]}{x * x} + \frac{X * P[X]}{X * X} - 0 \\
&= \mathcal{L}(x) + \mathcal{L}(X), \tag{35}
\end{aligned}$$

where we overloaded the notation of $\mathcal{L}$ to apply to any grade. Clearly this recurses over a blade composed of ℓ orthogonal vector factors to become the sum of the individual vector- $\mathcal{L}$ -functions. (Any invertible blade can be factorized in such an orthogonal manner by a Gram-Schmidt algorithm.) The k -round representing S^k is factorizable as $X = s \wedge \pi_1 \wedge \dots \wedge \pi_{n-k-1}$, with all factors having mutually zero dot products. We can split off the factors one by one and obtain

$$\begin{aligned}
& \mathcal{L}(s \wedge \pi_1 \wedge \dots \wedge \pi_{n-k-1}) \\
&= \mathcal{L}(s) + \mathcal{L}(\pi_1) + \dots + \mathcal{L}(\pi_{n-k-1}),
\end{aligned}$$

Therefore the cost function $\mathcal{L}(X)$ is the sum of the exact mean squared distances to each of the orthogonally intersecting planes π_i , plus the approximate mean squared distance to the sphere s in the sense of eq.(19). This is therefore indeed the extension to k -rounds of the cost criterion we have been using all along.⁷

6.2 The Best Fit as Pure Eigenproblem

Finding the extrema follows the same pattern we have seen before, and uses the identities eq.(16)-eq.(17):

$$\begin{aligned}
0 &= \partial_X \mathcal{L}(X) \\
&= (\partial_X (X * P[X])) (X * X)^{-1} \\
&\quad + (X * P[X]) \partial_X ((X * X)^{-1}) \\
&= 2P[X] (X * X)^{-1} - 2(X * P[X]) X (X * X)^{-2} \\
&= \frac{2}{X * X} \left(P[X] - \frac{X * P[X]}{X * X} X \right) \\
&= \frac{2}{X * X} \left(P[X] - \mathcal{L}(X) X \right).
\end{aligned}$$

⁷ For a 2-blade X , we compute $X * P[X] = \frac{1}{N} \sum_i X * (p_i \wedge (p_i \cdot X)) = -\frac{1}{N} \sum_i X * ((p_i \cdot X) \wedge p_i) = -\frac{1}{N} \sum_i ((p_i \cdot X) * (p_i \cdot X)) = -\frac{1}{N} \sum_i (p_i \cdot X)^2$, automatically providing the minus sign of eq.(25), as promised.

Therefore the eigen- ℓ -vectors of P provide extrema for $\mathcal{L}$, and the eigenvalue is the actually achieved cost. We are interested in the eigen- ℓ -blades. In an obvious extension of Lemma 1 using the property eq.(34), eigen- ℓ -blades can be constructed from the eigenvectors of the operator ‘ P applied to vectors’. Using Lemma 4, taking the first ℓ eigenvectors with nonnegative eigenvalues, a minimum k -sphere solving the fitting problem is $x_1 \wedge \dots \wedge x_\ell$. It is unique, unless the point data is accidentally distributed with some symmetry, such that some of the last of the first ℓ eigenvalues of P are identical to some of the first of the remaining eigenvalues (in which case one could use, say $x_{\ell+1}$ instead of x_ℓ in the optimal blade to form an equally optimal alternative solution).

6.3 Application: Fitting a Point Pair in 3D

In 3D, we have fitted a sphere S^2 in Section 3 and a circle S^1 in Section 5; the next order round is the point pair S^0 (‘a sphere along a line’). To compute it in a linear algebra package as the intersection of the best fit circle and the next best fit sphere, one needs to encode the outer product of a vector and a bivector to produce the trivector representing the point pair. Using the bivector basis $\{e_{23}, e_{31}, e_{12} \mid e_{o1}, e_{o2}, e_{o3} \mid e_{1\infty}, e_{2\infty}, e_{3\infty} \mid e_{o\infty}\}$ (as before) and a trivector basis $\{e_{1o\infty}, e_{2o\infty}, e_{3o\infty} \mid e_{o23}, e_{o31}, e_{o12} \mid e_{\infty23}, e_{\infty31}, e_{\infty12} \mid e_{123}\}$, the outer product matrix is:

$$[y \wedge X] = \begin{bmatrix} \mathbf{0} & -y_\infty [1] & -y_o [1] & \mathbf{y} \\ y_o [1] & -[\mathbf{y}^\times] & [0] & \mathbf{0} \\ y_\infty [1] & [0] & [\mathbf{y}^\times] & \mathbf{0} \\ \mathbf{y}^T & \mathbf{0}^T & \mathbf{0}^T & 0 \end{bmatrix} [X].$$

The resulting 10-component trivector then needs to be interpreted in terms of the 6 parameters of the point pair (p, q) . In CGA, this is straightforward using the formulas from Table 14.1 in [8]; in a linear algebra package without GA toolbox, one has to interpret those formulas in terms of the components of the trivector. Grouping the trivector components in order of the basis vectors as $(\mathbf{u}, \mathbf{v}, \mathbf{w}, \xi)$ (3 vectors and a scalar), developing a (dual) point pair $\alpha(p \wedge q)$ shows that $\mathbf{v} = \alpha(\mathbf{q} - \mathbf{p})$, that $\xi = \frac{1}{2}\alpha(\mathbf{q}^2 - \mathbf{p}^2) = \alpha(\frac{1}{2}(\mathbf{p} + \mathbf{q}) \cdot (\mathbf{q} - \mathbf{p}))$, and that $\mathbf{u} = \alpha(\mathbf{p} \times \mathbf{q}) = \alpha(\frac{1}{2}(\mathbf{p} + \mathbf{q}) \times (\mathbf{q} - \mathbf{p}))$. From this data, the vector $(\mathbf{p} + \mathbf{q})/2$ can be determined using the technique of eq.(33). The scale α may be computed from a geometrical invariant of the point pair trivector, like its squared radius $\rho^2 = -(\mathbf{u}^2 + 2\mathbf{v} \cdot \mathbf{w} - \xi^2)/\mathbf{v}^2$ (see Table 14.1 in [8]). This gives $\alpha = \frac{1}{2}\sqrt{\mathbf{v}^2/\rho^2}$ so that $\mathbf{q} - \mathbf{p} = \mathbf{v}/\alpha$. With the sum and difference of $\mathbf{p}$ and $\mathbf{q}$ thus determined, they can both be computed.

The resulting point pair is also indicated in Figure 6: it is the best point pair, whose mean of (approximate)

squared distances to the data points is minimized. For data generated from a sphere or circle, this is not a meaningful fit (it has a large residual cost); but for a bimodal distribution of points, this is a least squares method of retrieving its modes without the need for iterative separation into two classes.

7 Hyperaccurate Fits

In [1], various methods of fitting circles in 2D are compared, and their analysis extends fairly straightforwardly to hyperspheres in n -D. As we indicated before in Section 4, and as is confirmed in the simulation of Appendix A, all geometrically based algebraic fits are identical in their variance, but there is an essential bias which may manifest itself with enough data. The analysis allowed [1] to design a hyperaccurate fit method for circles in 2D, which has zero essential bias; we can easily extend it to n -D. We will use our CGA framework to encode this ‘hyperfit for hyperspheres’, and then discuss whether we can extend the hyperfit to k -spheres in n -D. Our work on the Pratt fit above will turn out to be very relevant!

The hyperfit as introduced in [1] is based on a modified constraint on the cost function

$$\text{cost: } \sum_i (A^2(x_i^2 + y_i^2) + Bx_i + Cy_i + D) \quad (36)$$

to fit a circle with algebraic parameters (A, B, C, D) to a set of data points with 2D coordinates (x_i, y_i) . Rather than enforcing a constraint by normalizing the squared radius $4A^2\rho^2 = B^2 + C^2 - 4AB$ (as Pratt does, see Section 4.1), they show that a hyperaccurate fit is one where the parameters (A, B, C, D) obey the constraint

$$8A^2\overline{x^2} + 8A^2\overline{y^2} + 4AB\overline{x} + 4AC\overline{y} + B^2 + C^2 + 4AD = 1, \quad (37)$$

see their equation (8.3), the overbars denote averages of coordinate expressions. Geometrically generalized to hyperspheres in their natural parameters of center $\mathbf{c}$ and radius ρ (rather than A, B, C, D), this is

$$\text{hyperfit: } 4A^2 (2(\|\bar{\mathbf{p}} - \mathbf{c}\|^2 + \sigma^2) - \rho^2) = 1,$$

where $\bar{\mathbf{p}}$ and σ are the centroid and standard deviation of the point data. Note that this criterion, as opposed to that of Pratt (and ours), therefore encodes some statistical properties of the data distribution.

For hypersphere fits, A is nonzero, and since it occurs as a factor both in cost function and constraint, we may divide it out. (In classical constrained optimization, this would eliminate the hyperplane solution

$A=0$, see [16]; but in the CGA approach, hyperplanes are still solutions when we divide cost and constraint.) We convert this criterion into CGA by introducing the *conformal centroid* $\bar{p}$:

$$\begin{aligned} \bar{p} &= \frac{1}{N} \sum_i p_i \\ &= \left(n_o + \frac{1}{N} \sum_i \mathbf{p}_i + \frac{1}{2} \left(\frac{1}{N} \sum_i \mathbf{p}_i^2 \right) n_\infty \right) \\ &= \left(n_o + \bar{\mathbf{p}} + \frac{1}{2} \bar{\mathbf{p}}^2 n_\infty \right) + \frac{1}{2} \left(\frac{1}{N} \sum_i \mathbf{p}_i^2 - \bar{\mathbf{p}}^2 \right) n_\infty \\ &= \text{CGA}(\bar{\mathbf{p}}) + \frac{1}{2} \sigma^2 n_\infty, \end{aligned}$$

where $\text{CGA}(\bar{\mathbf{p}})$ denotes the CGA representative of the centroid $\bar{\mathbf{p}}$ of the data (as in eq.(5)), and σ the standard deviation. The average of the conformal points is therefore an imaginary sphere with its center at the centroid $\bar{\mathbf{p}}$ of the data points, and a squared radius of minus their variance.

The hyperfit constraint in terms of CGA then takes the algebraically simple form:

$$2(\|\bar{\mathbf{p}} - \mathbf{c}\|^2 + \sigma^2) - \rho^2 = x^2 - 4(\bar{p} \cdot x). \quad (38)$$

Now introduce the rescaling function

$$R : \mathbb{R}^{n+1,1} \rightarrow \mathbb{R}^{n+1,1} : x \mapsto R[x] = x + 2(\bar{p} \cdot x) n_\infty.$$

R modifies a normalized sphere x with squared radius x^2 to one with squared radius $R[x]^2 = x^2 - 4(\bar{p} \cdot x)$, but the same center (hence the notation R , for *rescaling*). In terms of this rescaled sphere $R[x]$, the CGA hyperfit constraint reads $R[x]^2 = 1$.

7.1 Hyperfit hyperspheres

As we showed when treating the Pratt fit in Section 4.1, the cost criterion eq.(36) is written in CGA as $x \cdot P[x]$, with P as in eq.(22). We now observe that the quadratic function $x \cdot P[x]$ can be rephrased in terms of $R[x]$ as $R[x] \cdot P[R[x]]$. The computation is a straightforward rewriting when one realizes that

$$\bar{p} = \frac{1}{N} \sum_i p_i = \frac{1}{N} \sum_i p_i (-n_\infty \cdot p_i) = -P[n_\infty]. \quad (39)$$

We compute

$$\begin{aligned} R[x] \cdot P[R[x]] &= \\ &= (x + 2(\bar{p} \cdot x) n_\infty) \cdot P[x + 2(\bar{p} \cdot x) n_\infty] \\ &= (x + 2(\bar{p} \cdot x) n_\infty) \cdot (P[x] + 2(\bar{p} \cdot x) P[n_\infty]) \\ &= x \cdot P[x] + 2(\bar{p} \cdot x) (n_\infty \cdot P[x] + x \cdot P[n_\infty] + 2(\bar{p} \cdot x)) \\ &= x \cdot P[x] + 2(\bar{p} \cdot x) (2x \cdot P[n_\infty] + 2(\bar{p} \cdot x)) \\ &= x \cdot P[x]. \end{aligned}$$

Combining this with the constraint $R[x]^2 = 1$ incorporated as a denominator (as in Section 4.1), the hyper-accurate fit of a hypersphere is expressed in CGA as the x minimizing the function

$$\mathcal{L}_H(x) = \frac{R[x] \cdot P[R[x]]}{R[x]^2} = (R[x])^{-1} \cdot P[R[x]].$$

Comparing this with the cost function of eq.(23), we see immediately that $R[x]$ should be an eigenvector of P for the minimum non-negative eigenvalue. To retrieve x from $R[x]$, we observe that R is involutive:

$$\begin{aligned} R[R[x]] &= \\ &= x + 2(\bar{p} \cdot x) n_\infty + 2\left(\bar{p} \cdot (x + 2(\bar{p} \cdot x) n_\infty)\right) n_\infty \\ &= x + 4(\bar{p} \cdot x) n_\infty + 4(\bar{p} \cdot n_\infty)(\bar{p} \cdot x) n_\infty \\ &= x. \end{aligned}$$

We therefore find, in a small modification to the Pratt fit of Section 3.2:

The sphere $x_* = c - \frac{1}{2}\rho_H^2 n_\infty$ achieving the hyper-accurate total least squares fit to a set of conformal points $\{p_i\}$ is $x_* = y_* + 2\left(\left(\frac{1}{N} \sum_i p_i\right) \cdot y_*\right) n_\infty$, where y_* is the normalized eigenvector of minimum nonnegative eigenvalue of the linear operator $P[\cdot] = \frac{1}{N} \sum_i p_i (p_i \cdot [\cdot])$.

The above derivation shows that *the hyperfit sphere has the same center as the Pratt fit sphere* (though its radius usually differs). This has recently also been observed in [2].

The squared radius ρ_H^2 of the hyperfit hypersphere can be expressed in terms of the Pratt radius ρ and the radial variation of the data σ_{radial} , as follows. We first observe that the average sum of squared distances is an estimator for the radial variance σ_{radial}^2 , and that this is precisely $\mathcal{L}(y_*)$, the value of the cost function for the eigenvector y_* , which in turn equals the eigenvalue λ_* of P for a normalized y_* . We also observe that this eigenvalue is actually expressible as $\bar{p} \cdot y_*$, for using eq.(39) we have by the symmetry of P :

$$\bar{p} \cdot y_* = -P[n_\infty] \cdot y_* = -n_\infty \cdot P[y_*] = \lambda_* (-n_\infty \cdot y_*) = \lambda_*.$$

With those preliminaries, and under the usual assumption that σ_{radial} is small compared to ρ , we find:

$$\begin{aligned} \rho_H^2 &= x_*^2 = R[y_*]^2 = y_*^2 - 4(\bar{p} \cdot y_*) = \rho^2 - 4\lambda_* \\ &\approx \rho^2 - 4\sigma_{\text{radial}}^2 \approx (\rho - 2\sigma_{\text{radial}}/\rho)^2. \end{aligned} \quad (40)$$

The term $2\sigma_{\text{radial}}^2/\rho$ is precisely the bias of the Pratt radius given in [1] and also confirmed in our simulations of Appendix A. Subtracting it from the Pratt radius therefore indeed produces the radius of the hyperaccurate fit.

This computation works for hyperspheres of any dimension, although it is perhaps more natural to characterize the n -D data point noise by its total variance σ_{total}^2 ; then we should substitute $\sigma_{\text{radial}}^2 = \sigma_{\text{total}}^2/n$.

7.2 Towards hyperfit k -spheres

When computing the optimal k -spheres with the Pratt criterion, we lifted the function P from vectors representing spheres to blades representing k -spheres. Then we used the heredity of the optimization criterion under intersection of eq.(35) to factorize the problem recursively, and that led to the solution as intersection of eigenspheres of P .

Algebraically, we can treat the hyperfits similarly. It is easy to extend R from a sphere x to a blades X :

$$R[X] = X + 2n_\infty \wedge (\bar{p} \cdot X).$$

This is also involutive, i.e. $R[R[X]] = X$. And it is easy to verify by writing out that:

$$R[x \wedge X] = R[x] \wedge R[X].$$

But in deriving the counterpart to eq.(35) to factorize our problem, we face two problems. Firstly, even though the hyperaccurate hypersphere minimizes $\mathcal{L}(R[x])$, we do not know that the hyperfit k -blade minimizes $\mathcal{L}(R[X])$. The cost function, if it exists, should be derived from the general principles in [14]. Secondly, the hyperspheres $R[y]$ are radial rescalings of the orthogonal eigenspheres of P , and are no longer orthogonal (you may verify that $R[y_1] \cdot R[y_2] = 0$ does not follow from $y_1 \cdot y_2 = 0$). Even if $\mathcal{L}(R[X])$ would be the proper criterion, still its solution would not factorize and its minimization would therefore not be a straightforward transfer from the Pratt fit solution. For these reasons, we are obliged to leave the generalization from hyperfit hyperspheres to hyperfit k -spheres as future work.

8 Fitting Hyperflats

Hyperflats are translated hyperplanes and their intersections (hyperlines, etc.). They may occur as the outcome of a k -sphere fit if the data is degenerate, for instance lying in a plane in 3-D. Our method would provide these, since CGA has a ‘seamless’ (the term from [16]) representation for hyperspheres and hyperplanes. But if one knows *a priori* that the data has such a degeneracy, one may want to impose it, to avoid obtaining a hypersphere with very large radius as the best fit. We have performed these fits of hyperflats. Lack of space prevents us from giving the full details in this paper,

so we briefly indicate the essence of the method and its solution.

Since both hyperspheres and hyperplanes are vectors of CGA, the hypersphere fitting solution can be adapted to force-fit a hyperplane without too much effort. We only need to demand that the desired optimal hypersphere x should pass through the point at infinity, by imposing a Lagrange multiplier for the corresponding constraint $n_\infty \cdot x = 0$.

For a hyperplane x the cost function $\mathcal{L}(x) = x \cdot P[x]/x^2$ gives the *exact* mean square distance, so we obtain the same geometrical solution as in the classical approach. Algebraically, the CGA solution takes the form of an eigenproblem for the plane vector itself, rather than of an eigenproblem for the directional part of the position relative to the centroid.

Fitting hyperlines becomes: fitting hypercircles constrained to pass through infinity. Again the cost criterion for these flats is exact, and again an eigenproblem results. It has the same relationship to the eigenproblem for hyperplanes as we observed for hypercircles relative to hyperspheres. The best hyperline is the intersection of the two best fit perpendicular hyperplanes.

These results extend to fitting flats of any dimension. We are preparing a paper on the CGA representation of geometrical covariance of point data, which will contain the details of these solutions.

9 Conclusion

We have fit k -spheres to point data in n -D (with particular focus on spheres, circles and point-pairs in 3D). These fits were based on an approximate criterion that is algebraically convenient (effectively Pratt's criterion from [18]). We motivated the reasonability by experimentally confirming the theoretical results of [1], that for a practical amount of data sphere fits based on this criterion are virtually indistinguishable from an exact total least squares fit and their hyperaccurate fit. All these fits are optimal in the sense of achieving the theoretical lower bound of variance, and have a small or zero bias.

The algebraic convenience of the Pratt criterion is not obvious in the usual linear algebra formulation eq.(3). However, using the framework of conformal geometric algebra (CGA) where it takes the form eq.(4), we were able to derive a very effective relationship between hypersphere fitting and k -sphere fitting. We have shown how the hypersphere fitting problem leads to an eigenequation for a symmetric operator (our P of eq.(22)). This produces an orthogonal eigenbasis of perpendicularly intersecting hyperspheres. The eigenvector

with lowest eigenvalue corresponds to the best fit hypersphere.

The subspaces of the $(n+2)$ -D representational space in CGA represent k -spheres (and k -flats) of the Euclidean space it models. Subspaces can be combined by the outer product of their CGA vectors, which effectively computes the intersection of the corresponding Euclidean hyperspheres. When applied to the eigenbasis of P , the outer product of the two lowest eigenvectors directly produces the best fit hypercircle; the lowest $(n-k)$ produce the optimal k -spheres. We have demonstrated its general usefulness by showing how this procedure can be implemented in a standard linear algebra package (specifically, for 3D). Of course the computations would be simpler to specify and perform in a GA extension to such a package (such as [23]).

The cost criterion minimized by these fits is geometrically straightforward: if we consider a k -sphere as a hypersphere cut by a set of orthogonally intersecting hyperplanes, the cost criterion is the sum of exact squared distances to those hyperplanes plus the approximate squared distance to the hypersphere. Despite this interpretation, we have shown that the best fit circle is *not* the best fit sphere intersected by the best fit plane; and it is also *not* the best fit circle in the best fit plane. Instead, it is the best fit sphere intersected by the next best fit perpendicular sphere; intuitively the latter is the sphere that 'best takes care of the residue'.

After exposing this structural method based on the approximate least squares distances as encoded in the Pratt criterion, we tackled the hyperaccurate fit found in [1], which is not only optimal (like the Pratt fit, and some others), but also has zero essential bias. We proved that the hyperfit hypersphere is a radial rescaling of the Pratt fit hypersphere, relative to the same center.

To conclude, some general remarks about CGA. Though effective and compact in its treatment of k -sphere fitting, the CGA representation was not specifically designed just for this problem. In fact, CGA provides a structural, algebraic and coordinate-free way of computing with Euclidean geometry in general. For instance, CGA has been used (*avant-la-lettre*) in computational geometry to calculate n -D Voronoi diagrams by means of a convex hull in $(n+2)$ -D ([5], translated into CGA in [8]). CGA has a natural representation of rigid body motions (as spinors, an extension of quaternions). In this 'structure preserving' representation, all its geometrical primitives transform in exactly the same way. This convenient property enormously facilitates programming geometry [11,8]. CGA has been used in computer vision to estimate the parameters of placements from a variety of data (points, planes, lines) simultaneously [22], and for the interpolation of motions [9].

Should you be interested in learning more, a tutorial text written for a computer science audience is [8], which can serve as an entry to more mathematical treatments of geometric algebra like [13].

Acknowledgment

We thank an anonymous reviewer for encouraging us to investigate how to incorporate the hyperfit into CGA. For hyperspheres, this turned out to be a surprisingly minor modification, which led to our Section 7.

A Simulations of 3D Sphere Fitting

In the main text, we motivate our initial restriction to the Pratt fit for the extension to k -sphere by its quality relative to the geometric fit and the hyperaccuracy of the hyperfit of [1]. To substantiate the statements in [1] in this regard, we performed some simulations to compare the various fits for 3D spheres; some results are given in Figure 8 and 9. We used data points generated from a unit sphere, with variable radial Gaussian noise σ_{radial} (a standard deviation from 0.001 to 0.1, along the horizontal axis in each figure). Since one typically wants to use the fit to determine parameters of the sphere, we have characterized the properties of the fits by the average radius of the fit, and the standard deviation of this estimator. Both were determined by 50 repeated trials on 100 or 20 randomly generated data points (within every trial, the same random point set was used for all fitting methods). We plot two cases of directionality: an angular Gaussian distribution of the points with a standard deviation of 1 radian (Figure 8), and an angular Gaussian distribution with a standard deviation of 0.2 radians (Figure 9).

In the wide angular case of Figure 8, the various sphere fits all give good and stable estimates for the ground truth radius value of 1. The variance of these estimates are all very similar, confirming their equivalent ‘optimality’ [1] in achieving the KCR lower bound for their variance term. The differences in bias between Taubin fit [21], Pratt fit [18], geometric fit and hyperfit [1] completely confirm the theoretical results of [1] in units of $\sigma_{\text{radial}}^2/\rho$ being 2, 1, $\frac{1}{2}$, 0, respectively (which is a bit surprising in view of the considerable variance; the explanation is presumably the strong correlation between those fits, which we expose explicitly for Pratt fit and hyperfit in Section 7). For the 100 points used, the standard deviation exceeds the bias; other simulations (not shown) confirm our conclusion from Table 2 of [1] that only from about 1000 points onwards are these contributions to the MSE comparable.

In the narrow angular case of Figure 9 (fitting to a small spherical cap of points), as the noise increases the data point cloud becomes hard to distinguish from a cloud with a standard deviation of about 0.3, and the sphere fits ultimately all adapt to this. The algebraic fits of Coope [4] and Nievergelt [16], motivated on Frobenius norms of data matrices, do so for relatively low noise. The geometric fit (computed by Levenberg-Marquardt, with the Pratt fit as a seed) appears to fit very large radii around noise of 0.05, whereas the geometric/algebraic fits of Taubin, Pratt and the hyperfit behave somewhat better here.

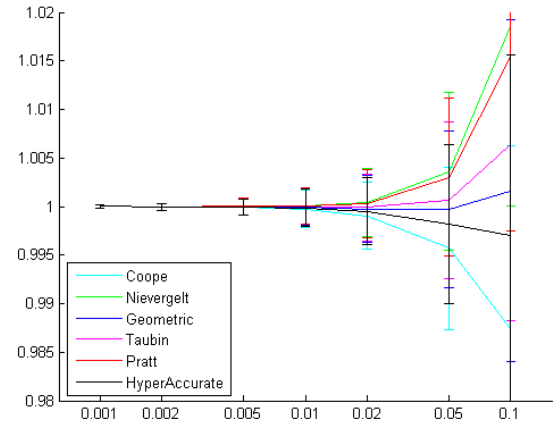


Fig. 8 Radius determination as a function of the radial noise standard deviation σ_{radial} , for spheres based on 100 data points, generated from a unit sphere, with angular standard deviation of 1 radian. With 50 trials per fit, we show average and standard deviation. Note the scale, all fits perform well.

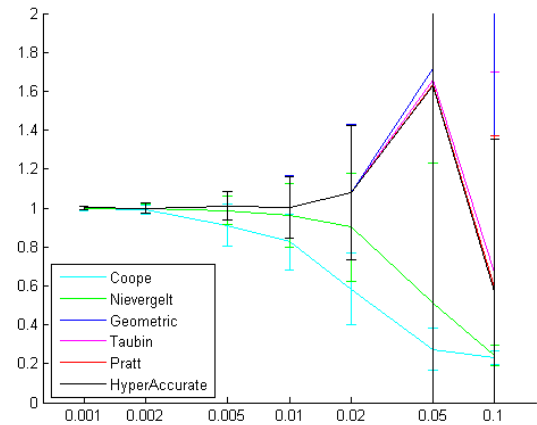


Fig. 9 Radius determination as a function of the radial noise standard deviation σ_{radial} , for spheres based on 20 data points, generated from a unit sphere, with angular standard deviation of 0.2 radian. With 50 trials per fit, we show average radius, and standard deviation.

These simulations confirm the basic results of [1] (and could have been done in that paper): that their hyperfit is best, that the geometrically ‘exact’ fit is biased, and that the bias of the Pratt fit is small, and only four times that of the geometric fit. We use this as a numerical motivation for focusing on the Pratt fit: it is very good as a fit, and it is algebraically preferable to the hyperfit for the desired extension to k -spheres.

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